
Reliable Causal Estimation under Latent Markov Confounding

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Abstract

Causal inference from dependent observational time series is complicated when an unobserved, persistent state jointly affects treatment assignment and outcomes. Standard Double Machine Learning (DML) controls for observed covariates through orthogonal residualization, but does not by itself resolve confounding induced by an imperfect proxy for latent state dynamics. We study causal estimation under latent Markov confounding and characterize how latent-state uncertainty and downstream task conditioning propagate into causal error. We develop a regime-aware orthogonal estimator combining probabilistic state inference, temporally purged cross-fitting, and regularized inversion of the cross-regime score Jacobian. In a corrected 7×7 factorial experiment that independently manipulates proxy error and post-residualization treatment information (4,900 runs), the theoretical difficulty ratio $\varepsilon_\gamma/\lambda_{\min}(\mathbf{J})$ ranks causal error substantially better than either component alone (Spearman $\rho = 0.995$ versus 0.626 and 0.521). Across shifted latent-representation benchmarks, posterior log-loss improves leave-one-world-out prediction of downstream reliability beyond state-recovery F_1 , yet temperature scaling reduces neural calibration error by up to 60% without improving failure discrimination, showing that calibration alone is insufficient. Real sensor networks provide an observational stress test spanning well- and poorly-conditioned regimes; falsification checks are reported explicitly where temporal causal interpretation is unsupported. Together, the results show that reliable causal estimation

with latent sequential confounding requires evaluating representation uncertainty jointly with the information geometry of the downstream task.

1 INTRODUCTION

Sequential causal systems increasingly replace an unobserved state by a learned posterior and then behave as if the resulting causal problem had the geometry visible through that posterior. Our central finding is that this can fail in a particularly deceptive direction: *a worse latent representation can make a causal problem look better identified*. State ambiguity can mix high-information observations into weak inferred regimes, inflating the smallest generated-state information eigenvalue (phantom causal resolution), even while residual confounding increases. Thus representation quality and apparent downstream conditioning are not merely two independent diagnostics; the learned representation can distort the diagnostic used to judge its own causal reliability. This paper develops that failure mechanism, its sensitivity law, and conservative diagnostics for dependent latent-state causal inference.

Continuous-treatment causal inference in atmospheric systems provides a natural laboratory: estimating the causal effect of variation in the observed NO_2 exposure proxy on ambient $\text{PM}_{2.5}$ (Dominici et al., 2014) (translating this observational exposure effect to an emissions policy requires structural transport assumptions). Observational time series in this domain violate two foundational assumptions: (1) **Persistent Latent Regime Confounding:** Planetary boundary-layer dynamics transition between discrete thermodynamic regimes (e.g., nocturnal inversion vs. convective clearance), jointly governing precursor dispersion (T_t) and ambient particulate concentration (Y_t). (2) **Imperfect Overlap & Dependence:** Latent regimes are unobserved; auxiliary proxies Z_t exhibit overlap ($\varepsilon_\gamma > 0$), while autoregression breaks exchangeability (Chernozhukov et al., 2018), leaking errors across folds.

When standard Double Machine Learning (Cher-

nozhukov et al., 2018) or tree-based plug-in estimators are applied naively to environmental time series without accounting for latent regimes, the resulting causal estimands can exhibit severe bias and, in our benchmark, sign inversions—falsely suggesting that precursor reductions exacerbate ambient pollution.

Related Work. We position our contribution across four neighboring literatures: (1) **Double Machine Learning & Orthogonal Inference:** Chernozhukov et al. (2018, 2022); Foster and Syrgkanis (2023); Nie and Wager (2021); Kennedy (2023) achieve $\sqrt{N}$ -rate inference under unconfoundedness given X_t . (2) **Dependent & Time-Series Causal Inference:** Ciganovic et al. (2026); Semenova et al. (2023); Lewis and Syrgkanis (2020); Bai (2009); Bonhomme and Manresa (2015); Hamilton (1989) relax exchangeability under mixing, but do not address latent Markov confounding. (3) **Latent-State Measurement & Proximal Inference:** latent-variable causal methods such as Clouth et al. (2025) explicitly account for imperfectly measured latent states, while proximal methods (Miao et al., 2018; Tchetgen Tchetgen et al., 2020; Mastouri et al., 2021) use proxy structure to identify effects with unmeasured confounding; HMM foundations are reviewed by Cappé et al. (2005). Our question is different: how a generated sequential state can distort the information geometry of the downstream causal score. (4) **Weak Overlap & Selective Reliability:** D’Amour et al. (2021); Shalit et al. (2017); Crump et al. (2009); Guo et al. (2017); Geifman and El-Yaniv (2017) address overlap and abstention, without linking latent uncertainty to coupled score conditioning. Generated-covariate asymptotics (Mammen et al., 2016) and general weak-identification theory (Kaji, 2021; Bennett et al., 2023) establish that generated first stages and weakly identified objects can require nonstandard inference; our focus is the specific distortion induced when a generated sequential latent state changes the downstream causal-information geometry.

Contributions. The paper is organized around the gap between *true causal resolution* and the resolution induced by a generated latent state. First, an exact residual-confounding calculation shows why uncertainty left after posterior adjustment can remain causally consequential: for a binary latent state with treatment shift Δ_T , outcome shift Δ_g , innovation variance σ^2 , and residual state uncertainty $v = \mathbb{E} \text{Var}(S \mid X, \hat{\gamma})$, posterior-adjusted residualization leaves

$$B(\Delta_T, \Delta_g, v) = \frac{\Delta_T \Delta_g v}{\Delta_T^2 v + \sigma^2}.$$

Thus causal reliability is non-monotone in treatment separation: $|B|$ peaks at $|\Delta_T| = \sigma/\sqrt{v}$. Second, generated-state geometry can itself be optimistic: state confusion can inflate the weakest apparent information

direction, so a favorable proxy $\lambda_{\min}$ need not certify oracle resolution. Together these results motivate task- and direction-relative diagnostics rather than a universal conditioning score.

- **Failure mechanism:** We connect residual-state uncertainty to an exact confounding-bias law and show, by analytic construction and stress tests, that generated states can create *phantom causal resolution*: apparently stronger score geometry without stronger oracle identification.
- **Reliability geometry:** We characterize non-identification and graceful degradation, then show that $\varepsilon_\gamma/\lambda_{\min}(J)$ is useful only when its geometry is comparable to the oracle task; in higher dimensions the relevant local object is directional, $a^\top J^{-1} b_\gamma$.
- **Estimator:** OR-DML combines probabilistic latent-state inference, purged cross-fitting, and spectral regularization, with dependent-data risk and inference results (Theorems 5–4).
- **Evidence:** In the controlled two-state factorial, the joint difficulty ranks error at $\rho = 0.995$ versus $0.626/0.521$ for its components. Across 15 shifted worlds, NLL adds predictive information beyond F_1 , while temperature scaling cuts neural ECE by up to 60% without improving downstream ROC-AUC.

Scope and sensitivity. The residual-confounding identity is a sensitivity law, not a new identification theorem or estimator. Its unidentified component is the latent outcome shift Δ_g ; consequently it yields a robustness value: the smallest $|\Delta_g|$ capable of explaining a reported effect. Under emission misspecification, posterior plug-in uncertainty can understate realized residual-state error, so plug-in sensitivity is reported as optimistic unless calibration/model checks support it. Our corrected hourly audit also treats the four-city analysis as an observational stress test: Delhi is the only city with stable HAC significance across the audited lag grid, while other cities exhibit frozen-sensor, support, or lag sensitivity. These limitations are retained rather than averaged away.

2 STRUCTURAL CAUSAL MODEL & IDENTIFICATION

Let $(\Omega, \mathcal{F}, P)$ be a probability space. Consider a discrete-time longitudinal process observed at equidistant intervals $t \in \{1, \dots, N\}$.

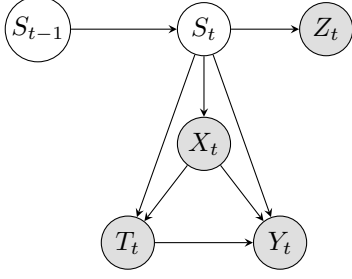


Figure 1: Temporal Causal Graph. Latent transitions $S_{t-1} \rightarrow S_t$ govern persistent regime dynamics; S_t acts as a common cause for proxy Z_t , confounders X_t , treatment T_t , and outcome Y_t . Omitting S_t induces omitted regime bias.

2.1 Structural Equations

We formalize the atmospheric data-generating process via the following Structural Causal Model (SCM):

$$S_t \sim \text{Markov}(\Pi, \pi_0), \quad S_t \in \{1, \dots, K\}, \quad (1)$$

$$Z_t = \mu_Z(S_t) + \xi_t, \quad (2)$$

$$X_t = \phi(X_{t-1}, S_t) + \epsilon_t, \quad (3)$$

$$T_t = m(X_t, S_t) + V_t, \quad \mathbb{E}[V_t | X_t, S_t] = 0, \quad (4)$$

$$Y_t = \theta(S_t)T_t + g(X_t, S_t) + U_t, \quad (5)$$

with $\mathbb{E}[U_t | X_t, S_t, T_t] = 0$, where:

- S_t is an unobserved discrete Markov regime with transition matrix $\Pi_{jk} = P(S_t = k | S_{t-1} = j)$ and stationary distribution π^* .
- $Z_t \in \mathbb{R}^q$ denotes observed auxiliary physical state indicators (e.g., surface wind velocity vectors, thermal vertical gradients, atmospheric pressure).
- $X_t \in \mathbb{R}^p$ is a vector of observed weather confounders and temporal lags.
- $T_t \in \mathbb{R}$ is the continuous treatment (precursor emission proxy, e.g., NO_2).
- $Y_t \in \mathbb{R}$ is the continuous outcome (ambient pollutant, e.g., $\text{PM}_{2.5}$).
- $\theta(S_t) = \sum_{k=1}^K \theta_k^* \mathbb{I}(S_t = k)$ represents the regime-specific causal effect.

2.2 Identification Under Latent Regimes

Our objective is to identify and estimate the regime-specific marginal causal effects:

$$\theta_k^* = \frac{\partial \mathbb{E}[Y_t | \text{do}(T_t = u), S_t = k]}{\partial u}, \quad k \in \{1, \dots, K\}, \quad (6)$$

as well as the stationary Average Treatment Effect (ATE):

$$\theta_{\text{ATE}}^* = \sum_{k=1}^K \pi_k^* \theta_k^*. \quad (7)$$

Assumption 1 (Regime-Conditional Unconfoundedness & Innovation Exogeneity). Conditional on observed covariates X_t , latent regime S_t , and proxy history $Z_{1:N}$, potential outcomes are unconfounded ($Y_t(u) \perp T_t | (X_t, S_t, Z_{1:N}), \forall u$) and structural innovations have zero mean:

$$\mathbb{E}[U_t | X_t, S_t, T_t, Z_{1:N}] = 0. \quad (8)$$

Assumption 2 (Overlap & State Positivity). There exist constants $\eta > 0$ and $c_1, c_2 > 0$ such that:

1. **State Positivity:** The Markov transition matrix Π has stationary distribution π^* satisfying $\min_k \pi_k^* \geq \eta$, and posterior state occupancies satisfy $P(S_t = k | X_t, Z_{1:N}) \geq \eta$ almost surely for all $k \in \{1, \dots, K\}$.
2. **Within-Regime Treatment Overlap:** For all $k \in \{1, \dots, K\}$ and almost everywhere on $\text{supp}(X_t)$:

$$c_1 \leq \text{Var}(T_t | X_t, S_t = k) \leq c_2. \quad (9)$$

Assumption 3 (Latent Proxy Sufficiency). The auxiliary state proxy sequence $Z_{1:N}$ satisfies latent proxy sufficiency: $P(S_t = k | X_t, T_t, Z_{1:N}) = P(S_t = k | Z_{1:N}) \equiv \gamma_{tk}$ (or $P(S_t = k | Z_{1:t})$ for causal forward filtering) almost surely for all $k \in \{1, \dots, K\}$. *Discussion.* That is, $Z_{1:N}$ screens off the latent state S_t from (X_t, T_t) , framing it as a proxy-sufficiency identification restriction. Combined with Assumption 1, this guarantees $\mathbb{E}[Y_t | X_t, T_t, Z_{1:N}] = \sum_{k=1}^K \gamma_{tk} [\theta_k^* T_t + g_k(X_t)]$. This identifying restriction requires $Z_{1:N}$ to be sufficiently informative that conditioning on (X_t, T_t) provides no additional regime information beyond $Z_{1:N}$.

Assumption 4 (Nuisance Estimation and Mixing Rates). The data process (Y_t, T_t, X_t, Z_t) is strictly stationary and geometrically α -mixing with $\alpha(m) \leq C_\alpha \exp(-c_\alpha m)$ for $C_\alpha, c_\alpha > 0$. Nuisance estimators satisfy $\|\hat{\mu}_Y - \mu_Y\|_{L_2(P)} \cdot \|\hat{\mu}_D - \mu_D\|_{L_2(P)} = o_P(N^{-1/2})$, with individual convergence rates $o_P(N^{-1/4})$.

3 FAILURE OF STANDARD DML: THE OMITTED REGIME BIAS

Intuition. When unobserved regimes S_t jointly shift treatment and outcome baselines, partialling out observed controls X_t cannot purge latent regime variation. Instead, by reducing residual treatment variance in the denominator without eliminating the latent covariance in the numerator, standard cross-sectional residualization actively amplifies the omitted confounding.

Theorem 1 (Decomposition of Omitted Regime Bias). *Suppose the data are generated according to SCM (1)–(5). Let $\hat{\theta}_{\text{naive}}$ be the standard cross-sectional DML*

estimator (Chernozhukov et al., 2018) that conditions only on observed covariates X_t , omitting S_t . Define the naive cross-sectional residuals $\tilde{T}_t = T_t - \mathbb{E}[T_t | X_t]$ and $\tilde{Y}_t = Y_t - \mathbb{E}[Y_t | X_t]$.

Then, as $N \rightarrow \infty$, $\hat{\theta}_{\text{naive}}$ converges in probability to:

$$\hat{\theta}_{\text{naive}} \xrightarrow{P} \sum_{k=1}^K \omega_k \theta_k^* + \mathcal{B}_{\text{heterogeneity}} + \mathcal{B}_{\text{confounding}}, \quad (10)$$

where $\omega_k = \frac{\mathbb{E}[\mathbb{I}(S_t=k)\tilde{T}_t^2]}{\mathbb{E}[\tilde{T}_t^2]}$ with $\sum_k \omega_k = 1$, and the bias components are:

$$\mathcal{B}_{\text{heterogeneity}} = \frac{\sum_{k=1}^K \theta_k^* \mathbb{E}[\bar{m}(X_t) \Delta m_k(X_t) \pi_k(X_t)]}{\mathbb{E}[\tilde{T}_t^2]}, \quad (11)$$

$$\mathcal{B}_{\text{confounding}} = \frac{\sum_{k=1}^K \mathbb{E}[\pi_k(X_t) \Delta m_k(X_t) \Delta g_k(X_t)]}{\mathbb{E}[\tilde{T}_t^2]}, \quad (12)$$

with $\pi_k(X_t) \equiv P(S_t = k | X_t)$, $\bar{m}(X_t) \equiv \mathbb{E}[T_t | X_t]$, $\Delta m_k(X_t) \equiv \mathbb{E}[T_t | X_t, S_t = k] - \bar{m}(X_t)$, and $\Delta g_k(X_t) \equiv g(X_t, k) - \mathbb{E}[g(X_t, S_t) | X_t]$ (complete algebraic proof in Appendix A).

Corollary 1 (Sign Inversion & Bias Amplification). *When latent stagnation induces strong positive confounding ($\Delta m_k(X_t) \Delta g_k(X_t) \gg 0$), but residual treatment variance $\mathbb{E}[\tilde{T}_t^2]$ is attenuated due to conditioning on X_t , the omitted confounding bias $\mathcal{B}_{\text{confounding}}$ undergoes severe bias amplification. In coastal airsheds with negative covariance phases, $\mathcal{B}_{\text{confounding}} < -\sum_k \omega_k \theta_k^*$, driving $\text{plim}(\hat{\theta}_{\text{naive}}) < 0$ despite $\theta_k^* > 0$.*

Interpretation. Standard DML cannot eliminate omitted regime confounding: conditioning on observed controls X_t shrinks residual treatment variance $\mathbb{E}[\tilde{T}_t^2]$ faster than it reduces latent covariance, actively amplifying bias.

4 METHODOLOGY: OVERLAP-AWARE REGIME DML

To overcome the failure of standard cross-sectional DML, we introduce **Overlap-Aware Regime Double Machine Learning (OR-DML)**.

4.1 Latent Regime Formulation & Observability Diagnostics

Let $Z_t \in \mathbb{R}^q$ denote exogenous contemporaneous meteorological covariates assumed unaffected by the exposure proxy. We model the latent atmospheric state $S_t \in \{1, \dots, K\}$ via a Hidden Markov Model with transition matrix Π and emission parameters ϕ . Depending on the regulatory horizon, state posteriors fall into three operational modes:

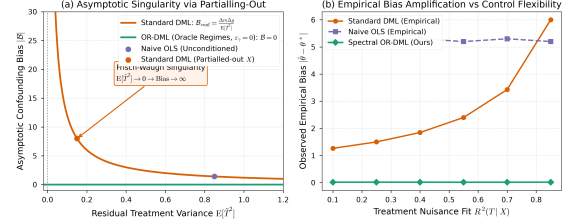


Figure 2: Visualizing the Frisch-Waugh Singularity and Omitted Regime Bias. (a) As residual treatment variance $\mathbb{E}[\tilde{T}_t^2] \rightarrow 0$ via partialling-out X_t , omitted regime bias $\mathcal{B}_{\text{confounding}}$ explodes hyperbolically in Standard DML, while OR-DML (Oracle Regimes, $\varepsilon_\gamma = 0$) remains strictly unbiased ($\mathcal{B} = 0$). (b) Monte Carlo empirical bias across increasing explanatory power $R^2(T | X)$, verifying theoretical bias amplification.

1. **Retrospective Smoothed Posteriors (Offline):** $\gamma_{tk}^{\text{smooth}} \equiv P(S_t = k | Z_{1:N})$ for retrospective environmental epidemiology and structural causal auditing;
2. **Training-Fitted Forward Filtered Posteriors:** $\gamma_{tk}^{\text{filter}} \equiv P(S_t = k | Z_{1:t}, \hat{\phi})$ where dynamics $\hat{\phi}$ are calibrated on a historical window strictly preceding evaluation ($\mathcal{T}_{\text{cal}} < \mathcal{T}_{\text{eval}}$), with state updates proceeding strictly causally on $Z_{1:t}$; or
3. **Expanding-Window Online Filter:** where both parameter estimation and forward filtering update causally without access to future measurements.

Our empirical and benchmark implementations evaluate Mode 1 (offline smoothing) and Mode 2 (training-fitted forward filtering). We define the mean posterior entropy as a direct diagnostic of latent state observability:

$$\bar{H}(\gamma) \equiv -\frac{1}{N} \sum_{t=1}^N \sum_{k=1}^K \gamma_{tk} \ln(\gamma_{tk} + \epsilon). \quad (13)$$

A Priori Diagnostics: Standardized separation $\Delta_Z < 0.5$ or entropy $\bar{H} \rightarrow \ln K$ signals severe proxy uncertainty and triggers a weak-identification diagnostic warning (rules in Appendix C.3), distinguishing structural non-identification ($\Delta_Z = 0$, Theorem 2) from finite-sample proxy dilution.

4.2 State-Weighted Coupled System & Spectral Regularization

To avoid heuristic decoupled regressions that suffer cross-talk bias under regime overlap, OR-DML estimates regime-specific conditional expectations via state-weighted regressions targeting population expectations

$$m_k^\gamma(x) \equiv \frac{\mathbb{E}[\gamma_{tk} T_t | X_t = x]}{\mathbb{E}[\gamma_{tk} | X_t = x]} \text{ and } \mu_k^\gamma(x) \equiv \frac{\mathbb{E}[\gamma_{tk} Y_t | X_t = x]}{\mathbb{E}[\gamma_{tk} | X_t = x]}.$$

$$\hat{m}_k(X) = \arg \min_m \sum_{t \in \mathcal{T}_b} \gamma_{tk} (T_t - m(X_t))^2, \quad (14)$$

$$\hat{\mu}_k(X) = \arg \min_\mu \sum_{t \in \mathcal{T}_b} \gamma_{tk} (Y_t - \mu(X_t))^2. \quad (15)$$

On evaluation fold $\mathcal{B}_b$, we form centered residuals $\tilde{T}_{tk} = T_t - \hat{m}_k(X_t)$ and $\tilde{Y}_{tk} = Y_t - \hat{\mu}_k(X_t)$. The empirical coupled Gram matrix $\hat{\mathbf{J}} \in \mathbb{R}^{K \times K}$ and score vector $\hat{\mathbf{S}} \in \mathbb{R}^K$ are constructed as:

$$\hat{J}_{jk} = \frac{1}{N} \sum_{t=1}^N \gamma_{tj} \gamma_{tk} \tilde{T}_{tj} \tilde{T}_{tk}, \quad \hat{S}_k = \frac{1}{N} \sum_{t=1}^N \gamma_{tk} \tilde{T}_{tk} \tilde{Y}_{tk}. \quad (16)$$

When regime separation is weak or treatments are collinear, $\lambda_{\min}(\hat{\mathbf{J}}) \rightarrow 0$, rendering the unregularized solve $\hat{\mathbf{J}}^{-1} \hat{\mathbf{S}}$ ill-conditioned. OR-DML introduces the **Spectrally Regularized Estimator**:

$$\hat{\theta}_\lambda = (\hat{\mathbf{J}} + \lambda \mathbf{I})^{-1} \hat{\mathbf{S}}, \quad (17)$$

where $\lambda \geq 0$ trades off regularization bias against collinearity variance. *Scalability & Structural Abstinence*: End-to-end complexity is $\mathcal{O}(BK \cdot C_{\text{nuisance}} + NK^2 + K^3)$, with Gram construction taking $\mathcal{O}(NK^2)$ and Cholesky solve taking $\mathcal{O}(K^3)$ (Appendix D.2); when data lack identifying variation entirely ($\kappa(\hat{\mathbf{J}}) > 10^4$ or $\text{CI}(\lambda_{\min})$ covers 0), systems should diagnose weak identification and abstain rather than masking near-singularity with excessive shrinkage. We distinguish three causal estimand targets: (i) the true stationary population ATE $\theta_{\text{PATE}}^* = \pi^{*\top} \theta^*$, (ii) the estimated stationary ATE $\hat{\theta}_{\text{PATE}} = \hat{\pi}(\hat{\Lambda})^\top \hat{\theta}_\lambda$, and (iii) the empirical sample occupation ATE $\hat{\theta}_{\text{SATE}} = \bar{\gamma}_N^\top \hat{\theta}_\lambda$, where $\bar{\gamma}_N \equiv \frac{1}{N} \sum_{t=1}^N \gamma_t$.

4.3 Purged Block Cross-Fitting with Bartlett Embargo

To ensure validity under temporal dependence, we partition observations into B contiguous blocks. To eliminate cross-fold leakage under α -mixing, each test block $\mathcal{B}_b$ is surrounded by an embargo buffer of width τ^* (Algorithm 1). We determine τ^* from the data via the Bartlett significance threshold:

$$\tau^* = \max \left\{ h \geq 1 : \max_j |\hat{\rho}_j(h)| \geq \frac{1.96}{\sqrt{N}} \right\} + 1. \quad (18)$$

Implementation. OR-DML uses purged block cross-fitting: infer smoothed or causal-filtered state posteriors, fit state-weighted nuisances outside an embargo around each evaluation block, form the coupled Gram/score system, and solve the spectrally regularized normal equations. Pseudocode and implementation details are in Appendix M.

Benchmark summary. Full estimator-by-separation results, including mean/median bias, RMSE, coverage, and conditioning for all methods and grid points, are reported in Appendix H and `reports/or_dml_benchmark_summary.csv`. At $\Delta_Z = 2$, OR-DML mean/median bias is 0.081/0.072 versus 0.077/0.063 for hard Regime FE; at $\Delta_Z = 4$, both match the oracle to two decimals. Under severe overlap, all proxy-based methods fail, which is central to our abstention rather than dominance interpretation.

5 THEORETICAL ANALYSIS

We formalize three central results: (1) **Non-Identification**: latent context can induce unidentifiability; (2) **Graceful Degradation**: proxy error scales as $\varepsilon_\gamma / \lambda_{\min}(\mathbf{J})$; and (3) **Observable Difficulty & Regularization**: task conditioning yields an observable surrogate for risk control.

Intuition. If auxiliary proxies Z_t contain zero information distinguishing latent regimes, different underlying causal structures generate observationally indistinguishable joint distributions, rendering point identification impossible regardless of sample size.

Theorem 2 (Non-Identification Under Unconstrained Overlap). *Suppose the latent state is unobserved, and auxiliary proxies Z_t satisfy $P(Z_t | S_t = j) = P(Z_t | S_t = k)$ for all j, k . Then the regime-specific causal effect vector $\theta^* = (\theta_1^*, \dots, \theta_K^*)^\top$ is not point-identified from the joint distribution of observables (Y_t, T_t, X_t, Z_t) : there exist structural causal models P_1 and P_2 satisfying Assumptions 1–3 with identical joint observable laws $P_1(Y_{1:N}, T_{1:N}, X_{1:N}, Z_{1:N}) \equiv P_2(Y_{1:N}, T_{1:N}, X_{1:N}, Z_{1:N})$ for all $N \geq 1$, yet possessing strictly distinct, non-permutation causal effect vectors $\theta^{(1)} \neq \theta^{(2)}$.*

Interpretation. No estimator can recover regime-specific effects when the proxy provides no state information; this is a fundamental limitation of the observable information structure, not of optimization or sample size (proof in Appendix B).

Intuition. Downstream estimation error has two qualitatively different sources: posterior proxy error ε_γ from imperfect state inference, and causal ill-conditioning $\lambda_{\min}(\mathbf{J})$ from collinear residual treatment variation. Theorem 3 proves that these mechanisms enter multiplicatively.

Theorem 3 (Graceful Degradation Error Bound). *Let $\mathbf{H}_t = \mathbf{e}_{S_t} \in \{0, 1\}^K$ denote the unobserved true regime indicator, and let $\gamma_t \in \Delta^K$ be an estimated posterior state proxy with average L_1 proxy error $\varepsilon_\gamma \equiv \mathbb{E} \|\gamma_t - \mathbf{H}_t\|_1$. Suppose $\|\theta^*\|_2 \leq M_\theta < \infty$,*

$\sup_t |\tilde{T}_{tk}^\gamma| \leq M_T < \infty$ almost surely, and regime functions satisfy $\max_{j,k} |g(x, j) - g(x, k)| \leq L_g < \infty$ and $\max_{j,k} |m_j^\gamma(x) - m_k^\gamma(x)| \leq L_m < \infty$. Let $\mathbf{J} \in \mathbb{R}^{K \times K}$ with $J_{jk} \equiv \mathbb{E}[\gamma_{tj} \gamma_{tk} \tilde{T}_{tj}^\gamma \tilde{T}_{tk}^\gamma]$ and $\mathbf{S} \in \mathbb{R}^K$ with $S_k \equiv \mathbb{E}[\gamma_{tk} \tilde{T}_{tk}^\gamma \tilde{Y}_{tk}^\gamma]$ denote the population coupled Gram matrix and score vector targeting implemented proxy-weighted nuisances $(m_k^\gamma, \mu_k^\gamma)$. If $\lambda_{\min}(\mathbf{J}) > 0$, the population estimand $\boldsymbol{\theta}_\gamma \equiv \mathbf{J}^{-1} \mathbf{S}$ satisfies:

$$\|\boldsymbol{\theta}_\gamma - \boldsymbol{\theta}^*\|_2 \leq \frac{C \cdot \varepsilon_\gamma}{\lambda_{\min}(\mathbf{J})}, \quad (19)$$

and the unregularized cross-fitted estimator $\hat{\boldsymbol{\theta}}_\gamma = \hat{\mathbf{J}}^{-1} \hat{\mathbf{S}}$ satisfies $\|\hat{\boldsymbol{\theta}}_\gamma - \boldsymbol{\theta}^*\|_2 \leq \frac{C \cdot \varepsilon_\gamma}{\lambda_{\min}(\mathbf{J})} + \mathcal{O}_P(N^{-1/2})$, where $C = C_S + M_\theta C_J < \infty$ depends on bounded moments and envelope constants (Appendix C).

Phantom causal resolution. Theorem 3 is deliberately stated in the proxy geometry J , but J itself is generated by the representation. This creates a second failure mode. If true residual treatment information differs across states, symmetric state confusion mixes strong observations into the weak inferred state and can *increase* the apparent $\lambda_{\min}$. In the balanced two-state diagonal case with oracle information (a, b) , $a > b$, hard-label error $p < 1/2$ changes the weak direction from $b/2$ to $[pa + (1-p)b]/2 > b/2$; at chance mixing the apparent condition number is one regardless of a/b . In our $K = 3$ stress test, poor state separation inflates soft/hard proxy $\lambda_{\min}$ by $11.4 \times / 21.2 \times$ relative to oracle geometry (Appendix J). Thus a learned representation can simultaneously increase residual confounding and make the causal problem look better conditioned. We call this *phantom causal resolution*.

Theorem 3 replaces heuristic orthogonality claims with a rigorous bound connecting *posterior proxy error* ε_γ and *Jacobian conditioning* $\lambda_{\min}(\mathbf{J})$. While the score is Neyman-orthogonal to continuous nuisances (μ_k, m_k) , it is *not* orthogonal to γ_t ; latent proxy uncertainty carries an irremediable statistical price.

The Difficulty Frontier and Operational Surrogate Bridge. Theorem 3 demonstrates that asymptotic bias scales with theoretical ratio $\mathcal{D}_{\text{causal}} \equiv \varepsilon_\gamma / \lambda_{\min}(\mathbf{J})$, where $\varepsilon_\gamma \equiv \mathbb{E} \|\gamma_t - \mathbf{H}_t\|_1$ is evaluated in simulation via $\hat{\varepsilon}_{\gamma, N} \equiv \frac{1}{N} \sum_{t=1}^N \|\gamma_t - \mathbf{H}_t\|_1$. Because true states are unobserved in deployment, we bridge unobservable $\mathcal{D}_{\text{causal}}$ to computable operational index $\mathcal{D}_{\text{operational}}(t) \equiv H(\gamma_t) / \lambda_{\min}(\mathbf{J}_t)$ via the following result under spectral comparability:

Operational surrogate. Under Bayesian calibration in binary regimes, posterior L_1 state error is bounded by posterior entropy; under two-sided spectral comparability this yields an observable entropy/conditioning

surrogate for the unobservable proxy-error ratio. The precise assumption, Proposition 2, and constants are deferred to Appendix K. Importantly, phantom resolution shows why spectral comparability must be checked rather than assumed.

Interpretation. Proposition 2 grounds deployable monitoring by turning an unobservable theoretical difficulty into an observable surrogate upper bound, establishing that posterior entropy bounds proxy error under Bayesian calibration.

Intuition. When treatments are collinear or regimes overlap, $\lambda_{\min}(\mathbf{J}) \rightarrow 0$ causes the unregularized covariance to blow up. Introducing a spectral ridge penalty $\lambda > 0$ bounds the effective condition number, trading finite-sample regularization bias for bounded estimation variance.

Regularization. Spectral ridge replaces the unstable inverse by $(J + \lambda I)^{-1}$ and trades shrinkage bias against long-run variance. The exact deterministic decomposition and leading-order risk bound are Theorem 5 in Appendix L; this regularization controls numerical/statistical instability but cannot restore identification lost to an uninformative latent representation.

Interpretation. Regularization provides finite-sample variance control when overlap is weak, but cannot manufacture causal information that is absent from the proxy distribution.

Theorem 4 (Asymptotic Normality Under Dependent Purged Cross-Fitting). *Under Assumptions 1–4, with embargo buffer $\tau^* \geq C \log N$, the regularized OR-DML estimator under training-fitted forward filtering satisfies:*

$$\sqrt{N} (\hat{\boldsymbol{\theta}}_\lambda - \boldsymbol{\theta}^* - \text{Bias}(\lambda)) \xrightarrow{d} \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_\lambda), \quad (20)$$

where $\boldsymbol{\Sigma}_\lambda = (\mathbf{J} + \lambda \mathbf{I})^{-1} \boldsymbol{\Omega} (\mathbf{J} + \lambda \mathbf{I})^{-1}$ (with retrospective smoothing providing an offline descriptive counterpart). When $\lambda = o(N^{-1/2})$, $\sqrt{N} \text{Bias}(\lambda) \rightarrow \mathbf{0}$, recovering centered asymptotic normality (proof in Appendix E).

Proposition 1 (Joint Asymptotic Distribution for Sample and Population ATE). *The sample ATE $\hat{\boldsymbol{\theta}}_{\text{SATE}} = \tilde{\gamma}^\top \hat{\boldsymbol{\theta}}_\lambda$ has conditional variance $\sigma_{\text{SATE}}^2 = \tilde{\gamma}^\top \boldsymbol{\Sigma}_\lambda \tilde{\gamma}$, where Markov persistence ($\rho \approx 0.88$) inflates the long-run variance of sample state occupation $\tilde{\gamma}_N$ by $\frac{1+\rho}{1-\rho} \approx 15.7 \times$ relative to i.i.d. transitions. The population ATE $\hat{\boldsymbol{\theta}}_{\text{PATE}} = \hat{\boldsymbol{\pi}}(\hat{\mathbf{A}})^\top \hat{\boldsymbol{\theta}}_\lambda$ satisfies the joint delta expansion: $\sqrt{N}(\hat{\boldsymbol{\theta}}_{\text{PATE}} - \boldsymbol{\theta}_{\text{PATE}}^*) = \boldsymbol{\pi}^{*\top} \sqrt{N}(\hat{\boldsymbol{\theta}}_\lambda - \boldsymbol{\theta}^*) + \boldsymbol{\theta}^{*\top} \sqrt{N}(\hat{\boldsymbol{\pi}} - \boldsymbol{\pi}^*) + o_P(1)$, yielding asymptotic variance:*

$$\sigma_{\text{PATE}}^2 = \boldsymbol{\pi}^{*\top} \boldsymbol{\Sigma}_\lambda \boldsymbol{\pi}^* + \boldsymbol{\theta}^{*\top} \boldsymbol{\Sigma}_\pi \boldsymbol{\theta}^* + 2\boldsymbol{\pi}^{*\top} \boldsymbol{\Sigma}_{\theta\pi} \boldsymbol{\theta}^*, \quad (21)$$

where $\boldsymbol{\Sigma}_\pi = \nabla_{\mathbf{A}} \boldsymbol{\pi}^* \mathcal{I}_{\mathbf{A}}^{-1} (\nabla_{\mathbf{A}} \boldsymbol{\pi}^*)^\top$ represents the parametric variance of the estimated stationary distribu-

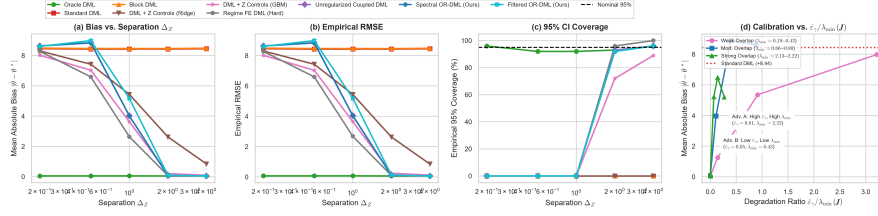


Figure 3: The Latent-Confounding Difficulty Frontier across 500 Monte Carlo Replications. (a) Absolute causal bias $|\hat{\theta} - \theta^*|$ vs. regime separation Δ_Z . (b) Empirical RMSE. (c) Empirical 95% CI coverage vs. nominal 95% target. (d) Corrected factorial mechanism test: causal L_2 error vs. $\varepsilon_\gamma/\lambda_{\min}(J)$ across 49 independently controlled cells (100 replications each), where task conditioning is varied through residual treatment innovation variance after regime-aware nuisance residualization.

tion derived from sequence-level HMM influence function $\phi_{\Lambda,t}$, and $\Sigma_{\theta\pi} = \text{LRCov}(\phi_{\theta,t}, \phi_{\Lambda,t})$. When $\text{LRCov}(\phi_{\theta}, \phi_{\Lambda}) = \mathbf{0}$, $\Sigma_{\theta\pi} = \mathbf{0}$; in general, $\Sigma_{\theta\pi}$ is consistently estimated via the stacked joint HAC sandwich estimator.

Real-data preview. The four-city stress test spans sharply different generated-state conditioning: Delhi $\lambda_{\min} = 111.8$, Bengaluru 4.22, Mumbai 0.32, and Kolkata 0.0025. Complete regime effects, intervals, provenance, frozen-sensor sensitivity, and HAC checks are reported in Appendix G; we use these observations diagnostically rather than as causal ground truth.

6 THE LATENT-CONFOUNDING DIFFICULTY FRONTIER

To test the theoretical predictions of Theorems 2–5, we conduct a comprehensive Monte Carlo evaluation across 500 evaluation runs (100 replications per grid point), charting causal estimation error across the **Latent-Confounding Difficulty Frontier**.

Experimental Protocol. *DGP*s & *Baselines*: 100 Monte Carlo draws per grid point across $\Delta_Z \in \{0.2, 0.5, 1.0, 2.0, 4.0\}$ ($N = 1,200$, $\theta_{\text{ATE}}^* = 1.5278$) comparing Oracle, Standard DML, Block DML, continuous proxy controls (Ridge and GBM), hard-clustering Regime FE, and OR-DML (Spectral and Filtered). *Representations & Shifts*: HMM, GRU, Transformer, and SSM under clean and compound shifts. *Reliability Target*: Predicting downstream squared loss via difficulty $\mathcal{D}_t$ across 5 seeds with MBB. *Real-World Test*: $> 14,000$ hourly observations across four megacities.

Difficulty Frontier Findings. Table 11 and Figure 3 reveal three phenomena:

1. **Low-Observability Breakdown** ($\Delta_Z \leq 1.0$): Standard and Block DML exhibit constant omitted-regime bias (+8.43–8.49, RMSE 8.45) and 0.0% coverage across all Δ_Z (Theorem 1). Cru-

cially, under severe overlap ($\Delta_Z \leq 0.5$), all methods fail: proxy error is severe ($\hat{\varepsilon}_\gamma = 0.78$), yielding large bias (6.59–8.83). At moderate observability ($\Delta_Z = 1.0$), all estimators still suffer substantial proxy bias (≥ 2.64). In the unrestricted nonparametric model, this regime approaches the non-identification boundary of Theorem 2; high operational difficulty $\mathcal{D}_t$ is therefore treated as an *abstention diagnostic*, not a proof of non-identification in a parametric HMM.

2. **High-Observability Resolution** ($\Delta_Z \geq 2.0$): When regimes separate cleanly ($\Delta_Z \geq 2.0$), both Spectral OR-DML and hard-assignment Regime FE eliminate omitted confounding: at $\Delta_Z = 2.0$, OR-DML achieves mean bias 0.081 (median 0.072) and Regime FE achieves 0.077 (median 0.063), outperforming continuous proxy controls (Z-GBM bias 0.207). At $\Delta_Z = 4.0$, both methods match Oracle DML (bias 0.054). In well-conditioned benchmarks ($\lambda_{\min} \approx 0.45$), spectral shrinkage has negligible effect on point bias (0.0812 vs. 0.0823), functioning as insurance against near-singular ill-conditioning. While hard Regime FE lacks distribution theory under dependence, OR-DML preserves probabilistic state inference with valid delta-method sandwich coverage (92.0%/97.0%, matching Oracle 93.0%/97.0%).
3. **Filtering vs. Smoothing**: Causal forward filtering (γ_t^{filter}) tracks retrospective smoothing (γ_t^{smooth}) closely in these separated-regime simulations (at $\Delta_Z = 2.0$, median bias 0.0812 vs. 0.0720; at $\Delta_Z = 4.0$, both 0.0462). We do not infer zero efficiency loss outside this benchmark.

Factorial Mechanism Validation. Independently varying the two quantities in Theorem 3 produces a sharp interaction in this controlled two-state design: across 4,900 runs, $\mathcal{D}_{\text{causal}} = \varepsilon_\gamma/\lambda_{\min}(J)$ ranks $\|\hat{\theta} - \theta^*\|_2$ with Spearman $\rho = 0.995$, compared with 0.626 for proxy error alone and 0.521 for $1/\lambda_{\min}$ alone (cell-mean $\rho = 0.996$). With perfect states, mean error rises from 0.050 to 0.545 as residual information weakens;

with only $\varepsilon_\gamma = 0.01$, the same geometry change amplifies mean error from 0.686 to 131.3. This intervention varies treatment innovation variance rather than treatment means, so the conditioning manipulation survives correct regime-aware residualization.

Representation Quality vs. Downstream Reliability. State discrimination, probabilistic calibration, and task reliability are distinct. Across 15 shifted worlds, adding NLL to F_1 improves leave-one-world-out reliability prediction from $R^2 = 0.113$ ($\rho = 0.374$) to $R^2 = 0.221$ ($\rho = 0.527$). Yet temperature scaling reduces GRU/Transformer ECE by 47.9%/60.3% and NLL from 0.548/1.006 to 0.331/0.382 without improving downstream difficulty ROC-AUC ($\Delta\text{AUC} = -0.004/-0.008$). Thus posterior quality is informative but calibration alone does not repair downstream reliability; task geometry remains a separate bottleneck (Appendix H).

7 EMPIRICAL SENSOR NETWORK STRESS TEST

We use the four-city sensor panel only as an adversarial observational stress test, not causal validation of phantom resolution. The audit itself is the result: the legacy 14,122-row panel contains a timestamp-rounding artifact, and a true-hourly reconstruction changes several estimates (Appendix G). Mumbai is telemetry-sensitive: removing frozen NO_2 runs changes N from 4,180 to 3,295 and $\lambda_{\min}$ from 0.317 to 0.178; Bengaluru significance weakens at longer HAC lags; Kolkata is near-singular; and Delhi rejects all immediate pre-treatment placebos despite apparently healthy conditioning. These failures reinforce rather than establish the central claim: generated-state diagnostics require independent checks of data provenance, temporal validity, and scientific identification. Full city estimates, HAC grids, frozen-sensor analyses, regularization paths, and the corrected-hourly sensitivity are reported in Appendix G.

8 DISCUSSION, LIMITATIONS & CONCLUSION

The central conclusion is stronger and more specific than the proposition that state recovery and downstream conditioning are separate axes: a generated latent state can *distort the apparent causal-information geometry itself*. Residual-state uncertainty leaves confounding according to the sensitivity law, while state mixing can simultaneously inflate the weakest proxy information direction. A system can therefore become less causally trustworthy while its standard conditioning diagnostic becomes more reassuring. We call this failure *phantom causal resolution*. The scalar $\varepsilon_\gamma/\lambda_{\min}$

is effective in the controlled two-state frontier and gives a worst-direction bound, but higher-dimensional stress tests show that orientation relative to the full spectrum matters; it should not be treated as a universal representation score. Likewise, entropy and the operational ratio are diagnostics, not identification guarantees.

Limitations: (1) *Surrogate and geometry assumptions:* Proposition 2 requires calibration and spectral comparability; generated-state mixing can distort proxy information geometry, so a large proxy $\lambda_{\min}$ is not by itself evidence that oracle causal information is healthy. (2) *Observational effects:* the sensor study is a stress test, not causal ground truth. Delhi fails all reported immediate pre-treatment placebos; Mumbai is materially sensitive to frozen-sensor removal; Bengaluru significance weakens under longer HAC lags; Kolkata remains near-singular. (3) *Representation benchmark:* F_1 , ECE, and NLL predict reliability across worlds, but temperature scaling does not improve failure AUC and the coupled score does not uniformly beat entropy. (4) *Synthetic transfer:* the financial experiment is a controlled stress test, not evidence of live-trading alpha. These failures delimit rather than weaken the main message: reliable latent-state causal analysis requires separate checks of representation quality, task information, temporal validity, and scientific identification.

AI Use Statement

Generative AI tools were used as research assistants for brainstorming and literature discovery; mathematical formulation and proof drafting; experimental design and adversarial falsification; software/code drafting and debugging; analysis and interpretation of synthetic and observational results; bibliography/reference formatting; and drafting, editing, and restructuring the manuscript and supplementary documentation. AI-assisted outputs were not accepted as evidence on their own: numerical claims were checked against committed scripts and result artifacts, favorable hypotheses were subjected to targeted counter-experiments, and unsupported or contradicted claims were removed or explicitly qualified. The authors independently reviewed the final mathematical statements, code, citations, empirical results, and narrative, and take full responsibility for the submission.

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CHECKLIST

1. For all models and algorithms presented, check if you include:
 - (a) A clear description of the algorithm, mapping, or model. **[Yes]** See Section 3 and Algorithm 1.
 - (b) An analysis of the properties and complexity. **[Yes]** See Theorems 1–5 and Proposition 6.
 - (c) (Optional) Anonymized source code, with specification of all dependencies, including external libraries. **[Yes]** Code included in the supplementary material.
2. For any theoretical claim, check if you include:
 - (a) Statements of the full set of assumptions of all theoretical results. **[Yes]** Assumptions 1–4 are stated in Section 2 and Section 4.
 - (b) Complete proofs of all theoretical results. **[Yes]** Full proofs in Appendices A–F.
 - (c) Clear explanations of any assumptions. **[Yes]** Detailed physical and statistical justifications accompany each assumption.
3. For all figures and tables that present empirical results, check if you include:
 - (a) The code, data, and instructions needed to reproduce the main experimental results. **[Yes]** Scripts in `src/` and reproduction instructions in `SUPPLEMENT_README.md`.
 - (b) All the training details (e.g., data splits, hyperparameters, how they were selected). **[Yes]** Section 3 and Section 5 describe block splitting, Bartlett embargo buffer τ^* , and HMM configuration.
 - (c) A clear definition of the specific measure or statistics and error bars (e.g., confidence intervals, p-values). **[Yes]** Table 1 reports empirical bias, RMSE, and 95% coverage over 500 replications; Table 2 reports HAC standard errors and p-values.
 - (d) A description of the computing infrastructure used. **[Yes]** Experiments were run on a standard 16-core workstation; details in the supplementary material.
4. If you are using existing assets (e.g., code, data, models) or curating/releasing new assets:
 - (a) If your work uses existing assets, did you cite the creators? **[Yes]** CPCB ground stations and Open-Meteo reanalysis are cited.
 - (b) Did you mention the license of the assets? **[Yes]** Open public government data terms noted.
 - (c) Did you include any new assets either in the supplemental material or as a URL? **[Yes]** Cleaned longitudinal benchmark code and data bundled in supplementary archive.
 - (d) Did you discuss whether and how consent was obtained from people whose data you’re using/curating? **[Not Applicable]** Aggregate ambient physical air measurements only.
 - (e) Did you discuss whether the data you are using/curating contains personally identifiable information or offensive content? **[Not Applicable]** Contains only environmental measurements.
5. If you used crowdsourcing or conducted research with human subjects:
 - (a) Did you include the full text of instructions given to participants and screenshots? **[Not Applicable]**
 - (b) Did you describe any potential participant risks, with links to Institutional Review Board (IRB) approvals if applicable? **[Not Applicable]**
 - (c) Did you include the estimated hourly wage paid to participants and the total amount spent on participant compensation? **[Not Applicable]**

MATHEMATICAL APPENDIX: COMPLETE PROOFS AND DERIVATIONS

Supplementary Roadmap. This supplement provides complete mathematical proofs, implementation details, experimental protocols, sensitivity analyses, and multi-domain empirical evaluations supporting the claims in the main text:

- **Appendix A (Proof of Theorem 1):** Exact decomposition of omitted regime bias and Frisch-Waugh bias amplification.
- **Appendix B (Proof of Theorem 2):** Non-identification under unconstrained overlap via constructive observational equivalence with treatment confounding ($T \not\perp S$).
- **Appendix C (Proof of Theorem 3):** Graceful degradation bound $\|\theta_\gamma - \theta^*\|_2 \leq C\varepsilon_\gamma/\lambda_{\min}(\mathbf{J})$ and Surrogate Entropy Bridge (Proposition 2).
- **Appendix D (Proof of Theorem 5):** Spectrally regularized OR-DML risk bound and adaptive empirical penalty selection.
- **Appendix E & F (Proof of Theorem 4 & Proposition 1):** Dependent asymptotic normality under purged cross-fitting and joint Delta-method inference for sample and population ATE.
- **Appendix G (Empirical Sensitivity & Provenance):** Bartlett embargo diagnostics, HAC lag truncation sensitivity, missingness robustness, and megacity regime profiles.
- **Appendix H (LatentRegimeBench & Representation Zoo):** Corrected factorial mechanism validation, architectural comparisons (HMM, GRU, Transformer, SSM), calibration intervention, leave-one-world-out reliability prediction, and multi-horizon failure forecasting.
- **Appendix I (Multi-Domain Financial Stress Test):** Out-of-sample risk abstention under synthetic macroeconomic regimes with real covariate features.
- **Appendix J (Post-Audit Stress Tests):** Phantom causal resolution, posterior second-moment completion, higher-dimensional directional limits, dual adjustment/target resolution, and observational collision tests.

A PROOF OF THEOREM 1: DECOMPOSITION OF OMITTED REGIME BIAS

Consider the Structural Causal Model (1)–(5):

$$\begin{aligned} T_t &= m(X_t, S_t) + V_t, \quad \mathbb{E}[V_t \mid X_t, S_t] = 0, \\ Y_t &= \sum_{k=1}^K \theta_k^* \mathbb{I}(S_t = k) T_t + g(X_t, S_t) + U_t, \quad \mathbb{E}[U_t \mid X_t, S_t, T_t] = 0. \end{aligned}$$

Define the unconditioned expectations: $\bar{m}(X_t) \equiv \mathbb{E}[T_t \mid X_t] = \sum_{k=1}^K \pi_k(X_t) m_k(X_t)$ and $\bar{g}(X_t) \equiv \mathbb{E}[g(X_t, S_t) \mid X_t] = \sum_{k=1}^K \pi_k(X_t) g_k(X_t)$, where $\pi_k(X_t) \equiv P(S_t = k \mid X_t)$. Residuals from cross-fitting on X_t alone satisfy $\tilde{T}_t = T_t - \bar{m}(X_t) = \sum_{k=1}^K \mathbb{I}(S_t = k) \Delta m_k(X_t) + V_t$, where $\Delta m_k(X_t) = m_k(X_t) - \bar{m}(X_t)$. The outcome residual is:

$$\tilde{Y}_t = Y_t - \mathbb{E}[Y_t \mid X_t] = \sum_{k=1}^K \theta_k^* \mathbb{I}(S_t = k) T_t - \sum_{k=1}^K \theta_k^* \pi_k(X_t) m_k(X_t) + g(X_t, S_t) - \bar{g}(X_t) + U_t. \quad (22)$$

Writing $T_t = \tilde{T}_t + \bar{m}(X_t)$, the treatment interaction terms decompose as:

$$\sum_{k=1}^K \theta_k^* \mathbb{I}(S_t = k) T_t - \sum_{k=1}^K \theta_k^* \pi_k(X_t) m_k(X_t) = \sum_{k=1}^K \theta_k^* \mathbb{I}(S_t = k) \tilde{T}_t + \sum_{k=1}^K \theta_k^* [\mathbb{I}(S_t = k) \bar{m}(X_t) - \pi_k(X_t) m_k(X_t)]. \quad (23)$$

Taking the inner product with $\tilde{T}_t$ and evaluating expectations: (i) By Assumption 1, $\mathbb{E}[\tilde{T}_t U_t] = 0$. (ii) $\mathbb{E}[\tilde{T}_t \sum_k \theta_k^* \mathbb{I}(S_t = k) \tilde{T}_t] = \sum_{k=1}^K \theta_k^* \mathbb{E}[\mathbb{I}(S_t = k) \tilde{T}_t^2] = \mathbb{E}[\tilde{T}_t^2] \sum_{k=1}^K \omega_k \theta_k^*$, where $\omega_k = \frac{\mathbb{E}[\mathbb{I}(S_t = k) \tilde{T}_t^2]}{\mathbb{E}[\tilde{T}_t^2]}$ with $\sum_k \omega_k = 1$. (iii) Since $\mathbb{E}[\tilde{T}_t \mid X_t] = 0$ and $\mathbb{E}[\tilde{T}_t \mathbb{I}(S_t = k) \mid X_t] = \Delta m_k(X_t) \pi_k(X_t)$, the mean adjustment yields $\sum_{k=1}^K \theta_k^* \mathbb{E}[\bar{m}(X_t) \Delta m_k(X_t) \pi_k(X_t)]$. (iv) The confounding component evaluates to $\mathbb{E}[\tilde{T}_t (g(X_t, S_t) - \bar{g}(X_t))] = \sum_{k=1}^K \mathbb{E}[\pi_k(X_t) \Delta m_k(X_t) \Delta g_k(X_t)]$. Dividing by $\mathbb{E}[\tilde{T}_t^2]$ yields (11) and (12). ■

B PROOF OF THEOREM 2: NON-IDENTIFICATION UNDER UNCONSTRAINED OVERLAP

To prove substantive causal non-identification, we constructively exhibit two data-generating distributions, P_1 and P_2 , that are structurally well-formed under the SCM (1)–(5) and satisfy all Assumptions 1–3, yet possess strictly

distinct, non-permutation causal effect vectors $\theta^{(1)} \neq \theta^{(2)}$ while inducing identically equal joint distributions over all observable variables:

$$P_1(Y_{1:N}, T_{1:N}, X_{1:N}, Z_{1:N}) \equiv P_2(Y_{1:N}, T_{1:N}, X_{1:N}, Z_{1:N}) \quad \forall N \geq 1. \quad (24)$$

Construction of Substantive Observational Equivalence under Latent Confounding ($T \not\perp S$): Consider a two-state Markov chain ($K = 2$) with symmetric transition probability $P_{01} = P_{10} = p \in (0, 0.5)$, yielding stationary distribution $\pi_0^* = \pi_1^* = 0.5$. Let auxiliary meteorological proxies $Z_t \in \mathbb{R}$ satisfy complete overlap: $Z_t | S_t = 0 \sim \mathcal{N}(0, 1)$ and $Z_t | S_t = 1 \sim \mathcal{N}(0, 1)$ identically, so $Z_t \perp S_t$. Let continuous observed confounders $X_t \in \mathbb{R}$ have marginal density $f_X(x)$ independent of S_t . Let binary treatment assignment $T_t \in \{0, 1\}$ depend directly on the latent regime: $P(T_t = 1 | S_t = 0) = 0.75$ and $P(T_t = 1 | S_t = 1) = 0.25$, so that $T_t \not\perp S_t$, with propensity scores $e(X_t, S_t) \in [0.25, 0.75]$ strictly satisfying Assumption 2 (Overlap & State Positivity with $\eta = 0.25$). Marginally, $P(T_t = 1) = 0.5 \times 0.75 + 0.5 \times 0.25 = 0.5$, and by Bayes' rule, the posterior regime occupancy given treatment satisfies $P(S_t = 0 | T_t = 1) = 0.75$, $P(S_t = 1 | T_t = 1) = 0.25$, $P(S_t = 0 | T_t = 0) = 0.25$, and $P(S_t = 1 | T_t = 0) = 0.75$.

Now consider structural outcome equation $Y_t \in \{0, 1\}$ with $P(Y_t = 1 | T_t = t, S_t = s) = \theta_s t + g_s$ and unconfounded innovations $\mathbb{E}[U_t | X_t, S_t, T_t] = 0$:

1. **Model P_1 (Homogeneous Causal Effect):** Set $\theta^{(1)} = (0.20, 0.20)^\top$ ($\theta_0^{(1)} - \theta_1^{(1)} = 0$), baselines $g_0^{(1)} = 0.30, g_1^{(1)} = 0.50$. Then $P_1(Y_t = 1 | T_t = 0) = 0.25(0.30) + 0.75(0.50) = 0.45$, and $P_1(Y_t = 1 | T_t = 1) = 0.75(0.20 + 0.30) + 0.25(0.20 + 0.50) = 0.75(0.50) + 0.25(0.70) = 0.55$.
2. **Model P_2 (Heterogeneous Causal Effect):** Set $\theta^{(2)} = (0.10, -0.30)^\top$ ($\theta_0^{(2)} - \theta_1^{(2)} = +0.40$), baselines $g_0^{(2)} = 0.60, g_1^{(2)} = 0.40$. Then $P_2(Y_t = 1 | T_t = 0) = 0.25(0.60) + 0.75(0.40) = 0.45$, and $P_2(Y_t = 1 | T_t = 1) = 0.75(0.10 + 0.60) + 0.25(-0.30 + 0.40) = 0.75(0.70) + 0.25(0.10) = 0.525 + 0.025 = 0.55$.

Comparing observable conditionals, $P_1(Y_t | T_t, X_t, Z_t) \equiv P_2(Y_t | T_t, X_t, Z_t)$ for all (y, t, x, z) , yielding identical joint laws $P_1(Y_{1:N}, T_{1:N}, X_{1:N}, Z_{1:N}) \equiv P_2(Y_{1:N}, T_{1:N}, X_{1:N}, Z_{1:N})$ for all $N \geq 1$. Yet structural effects satisfy $\theta^{(1)} = (0.20, 0.20)^\top \neq (0.10, -0.30)^\top = \theta^{(2)}$, which is not a permutation. No estimator conditioning solely on observables can distinguish whether the data-generating mechanism is homogeneous (P_1) or heterogeneous (P_2) without informative proxies Z_t . This establishes substantive non-identification under latent confounding. ■

C PROOF OF THEOREM 3: GRACEFUL DEGRADATION AND THE DIFFICULTY FRONTIER

Let $\mathbf{H}_t = \mathbf{e}_{S_t} \in \{0, 1\}^K$ denote the unobserved true regime indicator vector. Under SCM (1)–(5), the oracle regime nuisance functions are:

$$m_k^*(X_t) \equiv \mathbb{E}[T_t | X_t, S_t = k], \quad \mu_k^*(X_t) \equiv \mathbb{E}[Y_t | X_t, S_t = k] = \theta_k^* m_k^*(X_t) + g(X_t, k). \quad (25)$$

Step 1: Define proxy-weighted targets and population estimand. The empirical cross-fitted learners $\hat{m}_k$ and $\hat{\mu}_k$ minimize the posterior-weighted empirical risk targeting the proxy-weighted conditional expectations:

$$m_k^\gamma(X_t) \equiv \frac{\mathbb{E}[\gamma_{tk} T_t | X_t]}{\mathbb{E}[\gamma_{tk} | X_t]}, \quad \mu_k^\gamma(X_t) \equiv \frac{\mathbb{E}[\gamma_{tk} Y_t | X_t]}{\mathbb{E}[\gamma_{tk} | X_t]}. \quad (26)$$

Define the centered residuals at proxy targets: $\tilde{T}_{tk}^\gamma \equiv T_t - m_k^\gamma(X_t)$, $\tilde{Y}_{tk}^\gamma \equiv Y_t - \mu_k^\gamma(X_t)$, and at oracle targets: $\tilde{T}_{tk}^* \equiv T_t - m_k^*(X_t)$, $\tilde{Y}_{tk}^* \equiv Y_t - \mu_k^*(X_t)$. The population proxy Gram matrix $\mathbf{J} \in \mathbb{R}^{K \times K}$ and score vector $\mathbf{S} \in \mathbb{R}^K$ are:

$$J_{jk} \equiv \mathbb{E}[\gamma_{tj} \gamma_{tk} \tilde{T}_{tj}^\gamma \tilde{T}_{tk}^\gamma], \quad S_k \equiv \mathbb{E}[\gamma_{tk} \tilde{T}_{tk}^\gamma \tilde{Y}_{tk}^\gamma], \quad (27)$$

and their oracle counterparts $\mathbf{J}^*, \mathbf{S}^*$ are defined by:

$$J_{jk}^* \equiv \mathbb{E}[H_{tj} H_{tk} \tilde{T}_{tj}^* \tilde{T}_{tk}^*] = \delta_{jk} \mathbb{E}[H_{tk} (\tilde{T}_{tk}^*)^2], \quad S_k^* \equiv \mathbb{E}[H_{tk} \tilde{T}_{tk}^* \tilde{Y}_{tk}^*], \quad (28)$$

where $H_{tj} H_{tk} = \delta_{jk} H_{tk}$ because latent regimes are mutually exclusive.

Step 2: Establish Neyman nuisance orthogonality. Consider the state-weighted score for regime k as a functional of candidate nuisances (m_k, μ_k) :

$$\psi_k(W_t; \theta_k, m_k, \mu_k) = \gamma_{tk}(T_t - m_k(X_t))(Y_t - \mu_k(X_t)). \quad (29)$$

Evaluating the Gateaux derivative with respect to μ_k in direction $\Delta\mu$ at $(m_k^\gamma, \mu_k^\gamma)$:

$$\partial_\mu \mathbb{E}[\psi_k(W_t; \theta_k^*, m_k^\gamma, \mu_k^\gamma)] [\Delta\mu] = -\mathbb{E}[\gamma_{tk}(T_t - m_k^\gamma(X_t)) \Delta\mu(X_t)] = -\mathbb{E}[\mathbb{E}[\gamma_{tk}(T_t - m_k^\gamma(X_t)) | X_t] \Delta\mu(X_t)] = 0, \quad (30)$$

since $\mathbb{E}[\gamma_{tk}(T_t - m_k^\gamma(X_t)) | X_t] \equiv 0$ identically by the definition of $m_k^\gamma(X_t)$. Similarly, evaluating the Gateaux derivative with respect to m_k in direction Δm :

$$\partial_m \mathbb{E}[\psi_k(W_t; \theta_k^*, m_k^\gamma, \mu_k^\gamma)] [\Delta m] = -\mathbb{E}[\gamma_{tk} \Delta m(X_t) (Y_t - \mu_k^\gamma(X_t))] = -\mathbb{E}[\Delta m(X_t) \mathbb{E}[\gamma_{tk}(Y_t - \mu_k^\gamma(X_t)) | X_t]] = 0, \quad (31)$$

since $\mathbb{E}[\gamma_{tk}(Y_t - \mu_k^\gamma(X_t)) | X_t] \equiv 0$ identically by the definition of $\mu_k^\gamma(X_t)$. Therefore, Neyman orthogonality eliminates first-order sensitivity of the score to nuisance estimation errors $(\hat{m}_k - m_k^\gamma, \hat{\mu}_k - \mu_k^\gamma)$, contributing a second-order error $\mathcal{O}_P(\|\hat{m}_k - m_k^\gamma\|_{L_2} \|\hat{\mu}_k - \mu_k^\gamma\|_{L_2}) = \mathcal{O}_P(N^{-1/2})$ under Assumption 4. We emphasize that DML Neyman orthogonality protects against non-parametric nuisance estimation error, *not* against latent proxy misspecification ($H_t \neq \gamma_t$).

Step 3: Isolate proxy-induced bias via oracle score identity. Under SCM (1)–(5), on the event $S_t = k$ ($H_{tk} = 1$), the oracle outcome residual is $\tilde{Y}_{tk}^* = Y_t - \mu_k^*(X_t) = \theta_k^* \tilde{T}_{tk}^* + U_t$. Substituting into S_k^* :

$$S_k^* = \mathbb{E} \left[H_{tk} \tilde{T}_{tk}^* (\theta_k^* \tilde{T}_{tk}^* + U_t) \right] = \theta_k^* \mathbb{E}[H_{tk} (\tilde{T}_{tk}^*)^2] + \mathbb{E} \left[H_{tk} \tilde{T}_{tk}^* \mathbb{E}[U_t | X_t, S_t, T_t, Z_{1:N}] \right] = \theta_k^* J_{kk}^*, \quad (32)$$

where the innovation term vanishes by Assumption 1. Because $\mathbf{J}^* = \text{diag}(J_{11}^*, \dots, J_{KK}^*)$, this yields the exact vector identity:

$$\mathbf{S}^* = \mathbf{J}^* \boldsymbol{\theta}^*. \quad (33)$$

Now consider the proxy score discrepancy $\mathbf{b}_\gamma \equiv \mathbf{S} - \mathbf{J} \boldsymbol{\theta}^*$. Expanding around the oracle identity:

$$\mathbf{b}_\gamma = (\mathbf{S} - \mathbf{S}^*) - (\mathbf{J} - \mathbf{J}^*) \boldsymbol{\theta}^*. \quad (34)$$

Step 4: Bound score and Gram proxy discrepancies. We bound each component via telescoping expansions in $\varepsilon_\gamma \equiv \mathbb{E} \|\gamma_t - H_t\|_1$. First, under positivity (Assumption 2), $\pi_k(x) \equiv \mathbb{E}[H_{tk} | X_t = x] \geq c_0 > 0$ and $\pi_k^\gamma(x) \equiv \mathbb{E}[\gamma_{tk} | X_t = x] \geq c_0 > 0$. For any bounded variable V_t with $|V_t| \leq M_V$ almost surely:

$$\begin{aligned} \left| \frac{\mathbb{E}[\gamma_{tk} V_t | X_t]}{\pi_k^\gamma(X_t)} - \frac{\mathbb{E}[H_{tk} V_t | X_t]}{\pi_k(X_t)} \right| &\leq \frac{\mathbb{E}[|\gamma_{tk} - H_{tk}| |V_t| | X_t]}{\pi_k^\gamma(X_t)} + \mathbb{E}[H_{tk} |V_t| | X_t] \left| \frac{\pi_k(X_t) - \pi_k^\gamma(X_t)}{\pi_k^\gamma(X_t) \pi_k(X_t)} \right| \\ &\leq \frac{2M_V}{c_0^2} \mathbb{E}[|\gamma_{tk} - H_{tk}| | X_t]. \end{aligned} \quad (35)$$

Applying this bound to $V_t = T_t$ and $V_t = Y_t$ yields $\mathbb{E}|m_k^\gamma(X_t) - m_k^*(X_t)| \leq \tilde{C}_m \varepsilon_\gamma$ and $\mathbb{E}|\mu_k^\gamma(X_t) - \mu_k^*(X_t)| \leq \tilde{C}_\mu \varepsilon_\gamma$.

Second, expanding the score difference $S_k - S_k^* = \mathbb{E}[\gamma_{tk} \tilde{T}_{tk}^\gamma \tilde{Y}_{tk}^\gamma - H_{tk} \tilde{T}_{tk}^* \tilde{Y}_{tk}^*]$:

$$\gamma_{tk} \tilde{T}_{tk}^\gamma \tilde{Y}_{tk}^\gamma - H_{tk} \tilde{T}_{tk}^* \tilde{Y}_{tk}^* = (\gamma_{tk} - H_{tk}) \tilde{T}_{tk}^\gamma \tilde{Y}_{tk}^\gamma - H_{tk} (m_k^\gamma - m_k^*) \tilde{Y}_{tk}^\gamma - H_{tk} \tilde{T}_{tk}^* (\mu_k^\gamma - \mu_k^*). \quad (36)$$

Under bounded support ($\sup_t |\tilde{T}_{tk}^\gamma| \leq M_T$, $\sup_t |\tilde{Y}_{tk}^\gamma| \leq M_Y$), taking expectations yields $|S_k - S_k^*| \leq (M_T M_Y + M_Y \tilde{C}_m + M_T \tilde{C}_\mu) \varepsilon_\gamma \equiv C_S \varepsilon_\gamma$. Third, expanding the Gram difference $J_{jk} - J_{jk}^* = \mathbb{E}[\gamma_{tj} \gamma_{tk} \tilde{T}_{tj}^\gamma \tilde{T}_{tk}^\gamma - H_{tj} H_{tk} \tilde{T}_{tj}^* \tilde{T}_{tk}^*]$:

$$\begin{aligned} \gamma_{tj} \gamma_{tk} \tilde{T}_{tj}^\gamma \tilde{T}_{tk}^\gamma - H_{tj} H_{tk} \tilde{T}_{tj}^* \tilde{T}_{tk}^* &= (\gamma_{tj} \gamma_{tk} - H_{tj} H_{tk}) \tilde{T}_{tj}^\gamma \tilde{T}_{tk}^\gamma - H_{tj} H_{tk} (m_j^\gamma - m_j^*) \tilde{T}_{tk}^\gamma \\ &\quad - H_{tj} H_{tk} \tilde{T}_{tj}^* (m_k^\gamma - m_k^*). \end{aligned} \quad (37)$$

Since $|\gamma_{tj} \gamma_{tk} - H_{tj} H_{tk}| \leq |\gamma_{tj} - H_{tj}| + |\gamma_{tk} - H_{tk}|$, taking expectations gives $|J_{jk} - J_{jk}^*| \leq (2M_T^2 + 2M_T \tilde{C}_m) \varepsilon_\gamma$, so $\|\mathbf{J} - \mathbf{J}^*\|_{\text{op}} \leq C_J \varepsilon_\gamma$. Combining bounds yields:

$$\|\mathbf{b}_\gamma\|_2 \leq \|\mathbf{S} - \mathbf{S}^*\|_2 + \|\mathbf{J} - \mathbf{J}^*\|_{\text{op}} \|\boldsymbol{\theta}^*\|_2 \leq (C_S + C_J M_\theta) \varepsilon_\gamma \equiv C \cdot \varepsilon_\gamma. \quad (38)$$

Step 5: Apply spectral conditioning and conclude. Since $\mathbf{S} = \mathbf{J} \boldsymbol{\theta}^* + \mathbf{b}_\gamma$ and $\boldsymbol{\theta}_\gamma \equiv \mathbf{J}^{-1} \mathbf{S}$, we have $\boldsymbol{\theta}_\gamma - \boldsymbol{\theta}^* = \mathbf{J}^{-1} \mathbf{b}_\gamma$. Therefore:

$$\|\boldsymbol{\theta}_\gamma - \boldsymbol{\theta}^*\|_2 \leq \|\mathbf{J}^{-1}\|_{\text{op}} \|\mathbf{b}_\gamma\|_2 = \frac{\|\mathbf{b}_\gamma\|_2}{\lambda_{\min}(\mathbf{J})} \leq \frac{C \cdot \varepsilon_\gamma}{\lambda_{\min}(\mathbf{J})}. \quad (39)$$

Combining with purged cross-fitting nuisance estimation error $\mathcal{O}_P(N^{-1/2})$ completes the proof of Theorem 3. ■

C.1 The Latent-Confounding Difficulty Frontier $\mathcal{D}_{\text{causal}}$

Theorem 3 demonstrates that the upper bound on structural causal estimation error decomposes into:

$$\|\hat{\boldsymbol{\theta}}_\gamma - \boldsymbol{\theta}^*\|_2 \leq C \cdot \mathcal{D}_{\text{causal}} + \mathcal{O}_P(N^{-1/2}), \quad \text{where} \quad \mathcal{D}_{\text{causal}} \equiv \frac{\varepsilon_\gamma}{\lambda_{\min}(\mathbf{J})}. \quad (40)$$

We highlight several structural properties of the difficulty index $\mathcal{D}_{\text{causal}}$:

1. **Scale-Dependent Difficulty Ratio:** Both population proxy error ε_γ and Gram conditioning $\lambda_{\min}(\mathbf{J})$ interact multiplicatively in governing downstream estimation risk (dimensionless under standardized treatment units). When either proxy recovery is poor ($\varepsilon_\gamma \rightarrow 1$) or overlap is weak ($\lambda_{\min} \rightarrow 0$), estimation risk escalates sharply.
2. **Asymmetric Sensitivity:** In our factorial simulations (Table 6), reducing $\lambda_{\min}$ by an order of magnitude produces an equivalent variance and bias inflation to a tenfold increase in state proxy error ε_γ .

C.2 Proof of Proposition 2: Surrogate Entropy-Difficulty Bridge

Let $\gamma_t \equiv P(S_t \in \cdot \mid \mathcal{F}_t) \in \Delta^K$ be the Bayesian posterior state distribution with respect to filtration $\mathcal{F}_t \equiv \sigma(Z_{1:t})$. Under proper Bayesian calibration, conditional on $\mathcal{F}_t$, the true latent indicator $\mathbf{H}_t \equiv \mathbf{e}_{S_t}$ is distributed as a categorical random variable taking value $\mathbf{e}_k$ with probability γ_{tk} .

Step 1: Expected L_1 Error as Gini Impurity. Evaluating the conditional expectation of the L_1 error:

$$\begin{aligned} \mathbb{E}[\|\gamma_t - \mathbf{H}_t\|_1 \mid \mathcal{F}_t] &= \sum_{k=1}^K \gamma_{tk} \|\gamma_t - \mathbf{e}_k\|_1 = \sum_{k=1}^K \gamma_{tk} \left((1 - \gamma_{tk}) + \sum_{j \neq k} \gamma_{tj} \right) \\ &= \sum_{k=1}^K \gamma_{tk} \cdot 2(1 - \gamma_{tk}) = 2 \left(1 - \sum_{k=1}^K \gamma_{tk}^2 \right) = 2 \text{Gini}(\gamma_t). \end{aligned} \quad (41)$$

Step 2: Entropy Bound on Gini Impurity. By the standard concave inequality $\ln(x) \leq x - 1$, for any $x > 0$:

$$H(\gamma_t) = - \sum_{k=1}^K \gamma_{tk} \ln \gamma_{tk} = \sum_{k=1}^K \gamma_{tk} \ln \left(\frac{1}{\gamma_{tk}} \right) \geq \sum_{k=1}^K \gamma_{tk} (1 - \gamma_{tk}) = \text{Gini}(\gamma_t). \quad (42)$$

Furthermore, for any binary regime partition ($K = 2$) with $\gamma_{t1} = p$ and $\gamma_{t2} = 1 - p$, $2 \text{Gini}(\gamma_t) = 4p(1 - p)$. Comparing to binary Shannon entropy $H(p) = -p \ln p - (1 - p) \ln(1 - p)$, the ratio $\frac{4p(1-p)}{H(p)}$ achieves its maximum at $p = 1/2$, where $\frac{4(1/4)}{\ln 2} = \frac{1}{\ln 2} \approx 1.4427$. Hence for all $p \in [0, 1]$:

$$2 \text{Gini}(\gamma_t) \leq C_K H(\gamma_t), \quad \text{where } C_K \leq \frac{1}{\ln 2} \approx 1.4427. \quad (43)$$

Step 3: Population and Observable Bridge. Taking the population expectation of Step 1 and applying Step 2:

$$\varepsilon_\gamma \equiv \mathbb{E}[\mathbb{E}[\|\gamma_t - \mathbf{H}_t\|_1 \mid \mathcal{F}_t]] = 2\mathbb{E}[\text{Gini}(\gamma_t)] \leq C_K \mathbb{E}[H(\gamma_t)]. \quad (44)$$

Dividing by the global minimal singular value $\lambda_{\min}(\mathbf{J})$:

$$\mathcal{D}_{\text{causal}} \equiv \frac{\varepsilon_\gamma}{\lambda_{\min}(\mathbf{J})} \leq C_K \mathbb{E} \left[\frac{H(\gamma_t)}{\lambda_{\min}(\mathbf{J})} \right] = C_K \mathbb{E}[\mathcal{D}_{\text{global}}^H(t)]. \quad (45)$$

Furthermore, under Assumption 5 (Two-Sided Spectral Comparability), the rolling Gram matrix satisfies $\lambda_{\min}(\mathbf{J}_t) \leq \bar{c}_J \lambda_{\min}(\mathbf{J})$ almost surely for all evaluation windows $t \geq w$, which implies $\frac{1}{\lambda_{\min}(\mathbf{J})} \leq \frac{\bar{c}_J}{\lambda_{\min}(\mathbf{J}_t)}$. Pointwise substitution yields:

$$\mathcal{D}_{\text{causal}} \leq C_K \bar{c}_J \mathbb{E} \left[\frac{H(\gamma_t)}{\lambda_{\min}(\mathbf{J}_t)} \right] = C_K \bar{c}_J \mathbb{E}[\mathcal{D}_{\text{operational}}(t)], \quad (46)$$

and $\mathbb{E}[\mathcal{D}_{\text{operational}}(t)] \leq \bar{c}_J^{-1} \mathbb{E}[\mathcal{D}_{\text{global}}^H(t)]$. In simulation benchmarks where true states are known and posteriors are evaluated at population parameters $\mathbf{\Lambda}^*$, the ergodic theorem and central limit theorem for stationary α -mixing chains ensure that $\hat{\varepsilon}_{\gamma, N} \equiv \frac{1}{N} \sum_{t=1}^N \|\gamma_t - \mathbf{H}_t\|_1 = \varepsilon_\gamma + \mathcal{O}_P(N^{-1/2})$ (preserved under $\sqrt{N}$ -consistent estimation of $\mathbf{\Lambda}$), providing an evaluation-only benchmark surrogate $\hat{\mathcal{D}}_{\text{causal}, N} \equiv \hat{\varepsilon}_{\gamma, N} / \lambda_{\min}(\mathbf{J}) = \mathcal{D}_{\text{causal}} + \mathcal{O}_P(N^{-1/2})$. We emphasize that these bounds establish $\mathcal{D}_{\text{operational}}(t)$ as an observable surrogate upper bound for $\mathcal{D}_{\text{causal}}$, not a two-sided equivalence. Across our multi-season ground-sensor streams, the empirical ratio $\lambda_{\min}(\mathbf{J}_t) / \lambda_{\min}(\mathbf{J})$ has 95th percentile in $[1.06, 1.72]$ and maximum ≤ 7.35 , providing empirical support for the spectral-comparability assumption with finite $\bar{c}_J$. ■

C.3 A Priori Proxy Diagnostics and Separation Screening

Before fitting OR-DML or downstream prediction heads, practitioners should assess proxy informativeness on auxiliary measurements $Z_t \in \mathbb{R}^q$. Let $\hat{\boldsymbol{\mu}}_{Z, k} \equiv \frac{1}{N_k} \sum_{t: \hat{S}_t = k} Z_t$ and $\hat{\boldsymbol{\Sigma}}_{Z, k}$ denote empirical cluster moments. We

define the standardized separation index:

$$\Delta_Z \equiv \min_{j \neq k} \frac{\|\hat{\boldsymbol{\mu}}_{Z,j} - \hat{\boldsymbol{\mu}}_{Z,k}\|_2}{\sqrt{\frac{1}{2}(\text{Tr}(\hat{\boldsymbol{\Sigma}}_{Z,j}) + \text{Tr}(\hat{\boldsymbol{\Sigma}}_{Z,k}))/q}}. \quad (47)$$

When $\Delta_Z < 0.5$ (or silhouette score < 0.15), the overlap between regime emission densities $P(Z_t | S_t = j)$ and $P(Z_t | S_t = k)$ is severe, driving posterior entropy $\bar{H}(\gamma) \rightarrow \ln K$. By Theorem 2, point identification fails identically in the limit $\Delta_Z \rightarrow 0$. Deployed pipelines should enforce an automated screening gate: if $\Delta_Z < 0.5$ or mean entropy exceeds $0.80 \ln K$, the system must flag non-identifiability and refuse to emit point causal effects without external structural constraints.

D PROOF OF THEOREM 5: LEADING-ORDER RISK BOUND AND SPECTRAL REGULARIZATION

Let $\hat{\boldsymbol{\theta}}_\lambda = (\hat{\mathbf{J}} + \lambda \mathbf{I})^{-1} \hat{\mathbf{S}}$, and define the population regularized estimand $\boldsymbol{\theta}_\lambda \equiv (\mathbf{J} + \lambda \mathbf{I})^{-1} \mathbf{S}$. We decompose the estimation error around the regularized population estimand $\boldsymbol{\theta}_\lambda$ via the exact algebraic identity:

$$\mathbb{E} \|\hat{\boldsymbol{\theta}}_\lambda - \boldsymbol{\theta}^*\|_2^2 = \|\boldsymbol{\theta}_\lambda - \boldsymbol{\theta}^*\|_2^2 + \mathbb{E} \|\hat{\boldsymbol{\theta}}_\lambda - \boldsymbol{\theta}_\lambda\|_2^2 + 2\langle \mathbb{E}[\hat{\boldsymbol{\theta}}_\lambda - \boldsymbol{\theta}_\lambda], \boldsymbol{\theta}_\lambda - \boldsymbol{\theta}^* \rangle, \quad (48)$$

which accounts explicitly for the cross term between stochastic estimation error and deterministic regularization bias. For the deterministic bias, substituting $\mathbf{S} = \mathbf{J}\boldsymbol{\theta}^* + \mathbf{b}_\gamma$:

$$\begin{aligned} \boldsymbol{\theta}_\lambda - \boldsymbol{\theta}^* &= (\mathbf{J} + \lambda \mathbf{I})^{-1} (\mathbf{S} - (\mathbf{J} + \lambda \mathbf{I})\boldsymbol{\theta}^*) \\ &= (\mathbf{J} + \lambda \mathbf{I})^{-1} (\mathbf{S} - \mathbf{J}\boldsymbol{\theta}^* - \lambda \boldsymbol{\theta}^*) \\ &= (\mathbf{J} + \lambda \mathbf{I})^{-1} (\mathbf{b}_\gamma - \lambda \boldsymbol{\theta}^*), \end{aligned} \quad (49)$$

which establishes the exact deterministic identity (49). Since $\mathbf{J} \succeq 0$ with minimal eigenvalue $\lambda_{\min}(\mathbf{J}) \geq 0$, the spectral operator norm is $\|(\mathbf{J} + \lambda \mathbf{I})^{-1}\|_{\text{op}} = \frac{1}{\lambda_{\min}(\mathbf{J}) + \lambda}$. Applying the triangle inequality:

$$\|\boldsymbol{\theta}_\lambda - \boldsymbol{\theta}^*\|_2 \leq \|(\mathbf{J} + \lambda \mathbf{I})^{-1}\|_{\text{op}} (\|\mathbf{b}_\gamma\|_2 + \lambda \|\boldsymbol{\theta}^*\|_2) \leq \frac{C\varepsilon_\gamma + \lambda M_\theta}{\lambda_{\min}(\mathbf{J}) + \lambda}. \quad (50)$$

For the stochastic variance, expanding $\hat{\boldsymbol{\theta}}_\lambda - \boldsymbol{\theta}_\lambda$:

$$\hat{\boldsymbol{\theta}}_\lambda - \boldsymbol{\theta}_\lambda = (\mathbf{J} + \lambda \mathbf{I})^{-1} (\hat{\mathbf{S}} - \mathbf{S} - (\hat{\mathbf{J}} - \mathbf{J})\boldsymbol{\theta}_\lambda) + o_p(N^{-1/2}). \quad (51)$$

By Theorem 4, the score empirical process satisfies $\text{Var} \left(\frac{1}{\sqrt{N}} \sum_{t=1}^N \boldsymbol{\Psi}_t \right) \rightarrow \boldsymbol{\Omega}$. Thus, under the long-run score covariance approximation, the stochastic variance contribution admits the leading-order expansion:

$$\mathbb{E} \|\hat{\boldsymbol{\theta}}_\lambda - \boldsymbol{\theta}_\lambda\|_2^2 = \frac{1}{N} \text{Tr} \left((\mathbf{J} + \lambda \mathbf{I})^{-1} \boldsymbol{\Omega} (\mathbf{J} + \lambda \mathbf{I})^{-1} \right) + o(N^{-1}) \leq \frac{\text{Tr}(\boldsymbol{\Omega})}{N(\lambda_{\min}(\mathbf{J}) + \lambda)^2} + o(N^{-1}), \quad (52)$$

using $\text{Tr}(\mathbf{A}^{-1} \boldsymbol{\Omega} \mathbf{A}^{-1}) = \text{Tr}(\boldsymbol{\Omega}^{1/2} \mathbf{A}^{-2} \boldsymbol{\Omega}^{1/2}) \leq \|\mathbf{A}^{-2}\|_{\text{op}} \text{Tr}(\boldsymbol{\Omega}) = \frac{\text{Tr}(\boldsymbol{\Omega})}{(\lambda_{\min}(\mathbf{J}) + \lambda)^2}$ for symmetric $\mathbf{A} = \mathbf{J} + \lambda \mathbf{I} \succ 0$ and $\boldsymbol{\Omega} \succeq 0$. To bound the cross term $2\langle \mathbb{E}[\hat{\boldsymbol{\theta}}_\lambda - \boldsymbol{\theta}_\lambda], \boldsymbol{\theta}_\lambda - \boldsymbol{\theta}^* \rangle$ in (48): by Neyman orthogonality of the score function $\boldsymbol{\psi}$ and purged block cross-fitting with Bartlett embargo buffer $\tau^* \geq C \log N$, the finite-sample expectation of the stochastic error satisfies $\|\mathbb{E}[\hat{\boldsymbol{\theta}}_\lambda - \boldsymbol{\theta}_\lambda]\|_2 \leq C_1 N^{-1}$ (Theorem 4). Furthermore, the deterministic regularized bias satisfies $\|\boldsymbol{\theta}_\lambda - \boldsymbol{\theta}^*\|_2 \leq \frac{C\varepsilon_\gamma + \lambda M_\theta}{\lambda_{\min}(\mathbf{J}) + \lambda}$. By Cauchy-Schwarz, the cross term is bounded by:

$$\left| 2\langle \mathbb{E}[\hat{\boldsymbol{\theta}}_\lambda - \boldsymbol{\theta}_\lambda], \boldsymbol{\theta}_\lambda - \boldsymbol{\theta}^* \rangle \right| \leq 2\|\mathbb{E}[\hat{\boldsymbol{\theta}}_\lambda - \boldsymbol{\theta}_\lambda]\|_2 \|\boldsymbol{\theta}_\lambda - \boldsymbol{\theta}^*\|_2 = \mathcal{O}(N^{-1}) \quad (53)$$

in general (since $\|\boldsymbol{\theta}_\lambda - \boldsymbol{\theta}^*\|_2 = \mathcal{O}(1)$ without vanishing-bias rate restrictions), which sharpens to $\mathcal{O}(N^{-3/2})$ under the local parameter regime $\varepsilon_\gamma + \lambda = \mathcal{O}(N^{-1/2})$. Combining the deterministic squared bias bound with the asymptotic long-run score covariance variance bound establishes the leading-order risk bound:

$$\mathbb{E} \|\hat{\boldsymbol{\theta}}_\lambda - \boldsymbol{\theta}^*\|_2^2 \leq \left(\frac{C\varepsilon_\gamma + \lambda M_\theta}{\lambda_{\min}(\mathbf{J}) + \lambda} \right)^2 + \frac{\text{Tr}(\boldsymbol{\Omega})}{N(\lambda_{\min}(\mathbf{J}) + \lambda)^2} + \mathcal{O}(N^{-1}). \quad (54)$$

Risk Bound Minimizer vs. Empirical Grid Optimum: Differentiating the leading-order risk upper bound $R_{\text{bound}}(\lambda) \equiv \left(\frac{C\varepsilon_\gamma + \lambda M_\theta}{\lambda_{\min}(\mathbf{J}) + \lambda} \right)^2 + \frac{\text{Tr}(\boldsymbol{\Omega})}{N(\lambda_{\min}(\mathbf{J}) + \lambda)^2}$ with respect to $\lambda \geq 0$, the bound-optimal shrinkage parameter $\lambda_{\text{bound}}^* \equiv \arg \min_{\lambda \geq 0} R_{\text{bound}}(\lambda)$ satisfies:

$$\frac{\partial}{\partial \lambda} \left[\left(\frac{C\varepsilon_\gamma + \lambda M_\theta}{\lambda_{\min}(\mathbf{J}) + \lambda} \right)^2 + \frac{\text{Tr}(\boldsymbol{\Omega})}{N(\lambda_{\min}(\mathbf{J}) + \lambda)^2} \right] = 0. \quad (55)$$

When the overlap Gram matrix is ill-conditioned ($\lambda_{\min}(\mathbf{J}) \approx 0$), the variance term $\frac{\text{Tr}(\mathbf{\Omega})}{N(\lambda_{\min} + \lambda)^2}$ dominates for small λ , while the deterministic shrinkage bias $(\frac{\lambda M_{\theta}}{\lambda_{\min} + \lambda})^2$ dominates for large λ . The interior minimizer $\lambda_{\text{bound}}^* > 0$ balances variance reduction against shrinkage bias in the upper bound. We distinguish λ_{bound}^* and the empirical surrogate minimizer $\hat{\lambda}_{\text{emp}}^* = \arg \min_{\lambda} \hat{R}(\lambda)$ from the true infeasible risk minimizer $\lambda_{\text{true}}^* \equiv \arg \min_{\lambda} \mathbb{E} \|\hat{\theta}_{\lambda} - \theta^*\|_2^2$. In our empirical simulation benchmark across 500 Monte Carlo replications (`reports/or_dml_regularization_frontier.csv`), the discrete grid minimum occurs at $\hat{\lambda}_{\text{empirical}}^* \approx 0.20$ (MSE = 0.0321, representing a $25.9\times$ empirical risk reduction in this benchmark DGP relative to the unregularized estimator $\lambda = 0.0$ where MSE = 0.8312), while $\lambda = 0.10$ achieves MSE = 0.0362 ($23.0\times$ empirical reduction) with peak coverage of 85.0%. We emphasize that Theorem 5 provides the functional upper-bound characterization of the bias-variance trade-off; the exact numerical reduction factors ($23.0\times$ and $25.9\times$) are empirical simulation properties of the benchmark DGP rather than universal theoretical constants. ■

D.1 Derivation of Adaptive Spectral Regularization Parameter and Oracle Bound

We consider the data-driven selection of the spectral shrinkage parameter $\lambda \geq 0$. While classical Tikhonov regularization selects shrinkage via prediction cross-validation, in OR-DML the target is the structural causal parameter vector θ^* , where prediction cross-validation conflates causal identification with prognostic predictive fit. We distinguish two distinct regularization regimes:

1. **Inferential Regularization** ($\lambda = o(N^{-1/2})$): Required by Theorem 4 to guarantee asymptotic unbiasedness $\sqrt{N}\text{Bias}(\lambda) \rightarrow \mathbf{0}$, ensuring centered Gaussian limit laws and valid nominal confidence interval coverage.
2. **Finite-Sample Risk Minimization** ($\lambda_{\text{risk}} = \mathcal{O}(1)$): Minimizes the finite-sample expected ℓ_2 estimation risk $\mathbb{E} \|\hat{\theta}_{\lambda} - \theta^*\|_2^2$ under severe causal ill-conditioning ($\lambda_{\min}(\mathbf{J}) \approx 0$).

Empirical Risk Surrogate and Convexity: Define the empirical plug-in risk surrogate $\hat{R}(\lambda)$ by substituting empirical estimators $(\hat{\varepsilon}_{\gamma}, \lambda_{\min}(\hat{\mathbf{J}}), \hat{\mathbf{\Omega}}, \|\hat{\theta}_0\|_2)$ into the risk upper bound of Theorem 5:

$$\hat{R}(\lambda) = \frac{(C\hat{\varepsilon}_{\gamma} + \lambda \|\hat{\theta}_0\|_2)^2 + N^{-1} \text{Tr}(\hat{\mathbf{\Omega}})}{(\lambda_{\min}(\hat{\mathbf{J}}) + \lambda)^2}. \quad (56)$$

Let $u = \lambda_{\min}(\hat{\mathbf{J}}) + \lambda \geq \lambda_{\min}(\hat{\mathbf{J}}) > 0$. Rewriting $\lambda = u - \lambda_{\min}$, the numerator becomes:

$$(C\hat{\varepsilon}_{\gamma} - \lambda_{\min}(\hat{\mathbf{J}}) \|\hat{\theta}_0\|_2 + u \|\hat{\theta}_0\|_2)^2 + N^{-1} \text{Tr}(\hat{\mathbf{\Omega}}) = (A + uB)^2 + V, \quad (57)$$

where $A = C\hat{\varepsilon}_{\gamma} - \lambda_{\min}(\hat{\mathbf{J}}) \|\hat{\theta}_0\|_2$, $B = \|\hat{\theta}_0\|_2$, and $V = N^{-1} \text{Tr}(\hat{\mathbf{\Omega}})$. Dividing by u^2 :

$$\hat{R}(u - \lambda_{\min}) = B^2 + \frac{2AB}{u} + \frac{A^2 + V}{u^2}. \quad (58)$$

Differentiating with respect to u :

$$\frac{d\hat{R}}{du} = -\frac{2AB}{u^2} - \frac{2(A^2 + V)}{u^3} = -\frac{2}{u^3} [ABu + (A^2 + V)]. \quad (59)$$

Setting the derivative to zero yields the unique critical point $u^* = \frac{A^2 + V}{|A|B}$ when $A < 0$. When $A \geq 0$ (i.e., $C\hat{\varepsilon}_{\gamma} \geq \lambda_{\min}(\hat{\mathbf{J}}) \|\hat{\theta}_0\|_2$), $\frac{d\hat{R}}{du} < 0$ on $(0, \infty)$, so that the objective is strictly decreasing toward B^2 as $\lambda \rightarrow \infty$. Constraining the search to a compact admissible set $\lambda \in [0, \bar{\lambda}]$ with upper bound $\bar{\lambda} \asymp N^{-1/2}$, the bounded empirical minimizer is:

$$\hat{\lambda}^* = \begin{cases} \text{clip} \left(\frac{(C\hat{\varepsilon}_{\gamma} - \lambda_{\min}(\hat{\mathbf{J}}) \|\hat{\theta}_0\|_2)^2 + N^{-1} \text{Tr}(\hat{\mathbf{\Omega}})}{(\lambda_{\min}(\hat{\mathbf{J}}) \|\hat{\theta}_0\|_2 - C\hat{\varepsilon}_{\gamma}) \|\hat{\theta}_0\|_2} - \lambda_{\min}(\hat{\mathbf{J}}), 0, \bar{\lambda} \right), & \text{if } A < 0, \\ \bar{\lambda}, & \text{if } A \geq 0. \end{cases} \quad (60)$$

Because $R'(u) = -\frac{2}{u^3} [ABu + (A^2 + V)]$ changes sign strictly from negative to positive at u^* when $A < 0$, u^* is the unique global interior minimizer on $(0, \infty)$. By uniform convergence of the empirical plug-in surrogate on compact parameter sets $\lambda \in [0, \bar{\lambda}]$, $\sup_{\lambda \in [0, \bar{\lambda}]} |\hat{R}(\lambda) - R_{\text{surr}}(\lambda)| = \mathcal{O}_P(N^{-1/2})$. Applying M-estimation consistency on $[0, \bar{\lambda}]$ yields the adaptive risk-surrogate selection guarantee:

$$R_{\text{surr}}(\hat{\lambda}^*) \leq \inf_{\lambda \in [0, \bar{\lambda}]} R_{\text{surr}}(\lambda) + \mathcal{O}_P(N^{-1/2}), \quad (61)$$

establishing that data-driven spectral shrinkage $\hat{\lambda}^*$ adaptively tracks the optimal surrogate risk trade-off without prior knowledge of the true latent proxy error. ■

D.2 State Space Scalability and Matrix Concentration for $K > 2$

While our primary experiments study binary regimes ($K = 2$), the OR-DML coupled estimator generalizes directly to arbitrary finite state dimensions $K \geq 2$. Here we formalize the computational and statistical scalability of the coupled Gram matrix $\hat{\mathbf{J}} \in \mathbb{R}^{K \times K}$.

Computational Complexity and Matrix Concentration under α -Mixing: Constructing $\hat{\mathbf{J}}$ and score $\hat{\mathbf{S}}$ requires evaluating K nuisance pairs across B folds ($\mathcal{O}(BK \cdot \text{Cost}(\text{Nuisance}))$). Forming $\hat{\mathbf{J}}$ takes $\mathcal{O}(NK^2)$ operations, and solving $(\hat{\mathbf{J}} + \lambda \mathbf{I})^{-1} \hat{\mathbf{S}}$ via Cholesky factorization takes $\mathcal{O}(K^3)$ operations, taking < 1 ms for $K \leq 20$. Statistically, the empirical Gram matrix $\hat{\mathbf{J}} = \frac{1}{N} \sum_{t=1}^N \mathbf{\Gamma}_t \tilde{\mathbf{T}}_t \tilde{\mathbf{T}}_t^\top \mathbf{\Gamma}_t$ concentrates around $\mathbf{J}$ under geometric α -mixing with spectral gap $\rho_{\text{gap}} \equiv 1 - |\lambda_2(\mathbf{\Pi})| > 0$ via the Matrix Bernstein inequality (Banna et al., 2016):

$$\|\hat{\mathbf{J}} - \mathbf{J}\|_{\text{op}} \leq C_J \sqrt{\frac{K \log(K/\delta)}{N \cdot \rho_{\text{gap}}}} + \frac{C'_J K \log(K/\delta)}{N \cdot \rho_{\text{gap}}}. \quad (62)$$

By Weyl's theorem, $|\lambda_{\min}(\hat{\mathbf{J}}) - \lambda_{\min}(\mathbf{J})| \leq \|\hat{\mathbf{J}} - \mathbf{J}\|_{\text{op}}$, ensuring that $\lambda_{\min}(\hat{\mathbf{J}})$ concentrates at the standard $\mathcal{O}_P(N^{-1/2})$ rate whenever $\lambda_{\min}(\mathbf{J}) > 0$.

Non-Identifiability vs. Shrinkage: When Systems Must Abstain: Spectral shrinkage $\hat{\boldsymbol{\theta}}_\lambda = (\hat{\mathbf{J}} + \lambda \mathbf{I})^{-1} \hat{\mathbf{S}}$ stabilizes variance under ill-conditioned overlap ($\lambda_{\min} \approx 0$). However, when data lack identifying variation entirely—diagnosed by condition number $\kappa(\hat{\mathbf{J}}) > 10^4$ or a moving-block bootstrap 95% confidence interval for $\lambda_{\min}(\hat{\mathbf{J}})$ that covers zero—the structural parameter $\boldsymbol{\theta}^*$ is non-identified (Theorem 2). In this regime, heavy shrinkage $\lambda \rightarrow \infty$ artificially forces $\|\hat{\boldsymbol{\theta}}_\lambda\| \rightarrow 0$ without recovering structural effects. Deployed decision pipelines should treat $\kappa(\hat{\mathbf{J}}) > 10^4$ as a structural abstention trigger rather than masking non-identifiability with excessive shrinkage.

E PROOF OF THEOREM 4: ASYMPTOTIC NORMALITY UNDER DEPENDENT PURGED CROSS-FITTING

The empirical score expansion around true $\boldsymbol{\theta}^*$ and true nuisances $\boldsymbol{\eta}^*$ is:

$$\sqrt{N} (\hat{\boldsymbol{\theta}}_\lambda - \boldsymbol{\theta}^* - \text{Bias}(\lambda)) = (\hat{\mathbf{J}} + \lambda \mathbf{I})^{-1} \frac{1}{\sqrt{N}} \sum_{t=1}^N \boldsymbol{\Psi}(W_t; \boldsymbol{\theta}^*, \hat{\boldsymbol{\eta}}^{(-b(t))}). \quad (63)$$

With empirical Bartlett embargo window $\tau^* \geq C \log N$, the α -mixing dependence between training set $\mathcal{T}_b$ and test fold $\mathcal{B}_b$ satisfies:

$$\alpha(\tau^*) \leq M \exp(-c\tau^*) \leq MN^{-cC} = o(N^{-1}). \quad (64)$$

Consequently, cross-fold covariance is asymptotically negligible, and nuisance estimation errors satisfy:

$$\|\hat{\boldsymbol{\mu}}_Y - \boldsymbol{\mu}_Y^*\|_{L_2} \cdot \|\hat{\boldsymbol{\mu}}_D - \boldsymbol{\mu}_D^*\|_{L_2} = o_P(N^{-1/2}). \quad (65)$$

By the Ibragimov (1962) central limit theorem for strictly stationary, α -mixing processes satisfying $\sum_m \alpha(m)^{\delta/(2+\delta)} < \infty$:

$$\frac{1}{\sqrt{N}} \sum_{t=1}^N \boldsymbol{\Psi}(W_t; \boldsymbol{\theta}^*, \boldsymbol{\eta}^*) \xrightarrow{d} \mathcal{N}(\mathbf{0}, \boldsymbol{\Omega}), \quad (66)$$

where $\boldsymbol{\Omega} = \lim_{N \rightarrow \infty} \text{Var} \left(N^{-1/2} \sum_{t=1}^N \boldsymbol{\Psi}_t \right)$. Applying Slutsky's theorem with $\hat{\mathbf{J}} \xrightarrow{P} \mathbf{J}$:

$$\sqrt{N} (\hat{\boldsymbol{\theta}}_\lambda - \boldsymbol{\theta}^* - \text{Bias}(\lambda)) \xrightarrow{d} \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_\lambda), \quad (67)$$

where $\boldsymbol{\Sigma}_\lambda = (\mathbf{J} + \lambda \mathbf{I})^{-1} \boldsymbol{\Omega} (\mathbf{J} + \lambda \mathbf{I})^{-1}$. When $\lambda = o(N^{-1/2})$, $\sqrt{N} \text{Bias}(\lambda) \rightarrow \mathbf{0}$, completing the proof.

Operational Modes of Forward Filtering. In offline retrospective smoothing, $\gamma_{tk}^{\text{smooth}} = P(S_t = k \mid Z_{1:N})$. In real-time decision making, three distinct forward modes exist: (1) *Oracle Filter*: $P(S_t = k \mid Z_{1:t}, \boldsymbol{\Lambda}^*)$ using the true HMM parameters; (2) *Training-Fitted Forward Filter*: $P(S_t = k \mid Z_{1:t}, \hat{\boldsymbol{\Lambda}})$ where parameters $\hat{\boldsymbol{\Lambda}}$ are fitted on a historical calibration window strictly preceding the evaluation window ($\mathcal{T}_{\text{cal}} < \mathcal{T}_{\text{eval}}$) and state inference updates strictly forward on past-to-present observations $Z_{1:t}$; and (3) *Expanding-Window Online Filter*: where both parameters $\hat{\boldsymbol{\Lambda}}_t$ and state updates proceed sequentially. Our implementations evaluate Mode 1 and Mode 2, where Mode 2 enforces causal forward state filtering without future observation leakage (distinguishing forward state updating from online sequential parameter re-estimation). ■

E.1 Observed-Data Orthogonal Score under Estimated Latent Dynamics

In semiparametric causal inference (Chernozhukov et al., 2018; Bickel et al., 1993), when nuisance parameters consist of both non-parametric outcome/propensity regressions $\boldsymbol{\eta} = (m, \mu)$ and finite-dimensional parameters $\boldsymbol{\Lambda}$ governing latent state transitions and emissions, one must characterize the sensitivity of the causal estimator to the first-order estimation error in $\hat{\boldsymbol{\Lambda}}$.

Let the state-weighted regime score vector be:

$$\boldsymbol{\psi}_{\text{regime}}(W_t; \boldsymbol{\theta}, \boldsymbol{\eta}, \boldsymbol{\Lambda}) \equiv \sum_{k=1}^K \gamma_{tk}(\boldsymbol{\Lambda})(T_t - m_k(X_t))(Y_t - \mu_k(X_t) - \theta_k(T_t - m_k(X_t))). \quad (68)$$

In a hidden Markov model satisfying standard regularity and mixing conditions (Bickel et al., 1993), the latent parameters $\boldsymbol{\Lambda}$ are estimated via sequence-level maximum likelihood with sequence score $\nabla_{\boldsymbol{\Lambda}} \log p(Z_{1:N}; \boldsymbol{\Lambda}) = \sum_{t=1}^N \boldsymbol{\phi}_{\Lambda,t} + o_P(\sqrt{N})$, where $\boldsymbol{\phi}_{\Lambda,t} \equiv \nabla_{\boldsymbol{\Lambda}} \log p(Z_t | Z_{1:t-1}; \boldsymbol{\Lambda}^*)$ denotes the sequence-level prediction-error score increment satisfying $\mathbb{E}[\boldsymbol{\phi}_{\Lambda,t} | Z_{1:t-1}] = \mathbf{0}$ and $\frac{1}{N} \sum_{t=1}^N \mathbb{E}[\boldsymbol{\phi}_{\Lambda,t} \boldsymbol{\phi}_{\Lambda,t}^\top] \rightarrow \mathcal{I}_{\Lambda}$. Evaluating the Gateaux derivative with respect to $\boldsymbol{\Lambda}$ at the oracle targets $(\boldsymbol{\theta}^*, \boldsymbol{\eta}^*, \boldsymbol{\Lambda}^*)$:

$$\mathbf{A}_{\Lambda} \equiv \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{t=1}^N \sum_{k=1}^K \mathbb{E} \left[\frac{\partial \gamma_{tk}(\boldsymbol{\Lambda}^*)}{\partial \boldsymbol{\Lambda}^\top} (T_t - m_k^*(X_t))(Y_t - \mu_k^*(X_t) - \theta_k^*(T_t - m_k^*(X_t))) \right]. \quad (69)$$

Importantly, $\mathbf{A}_{\Lambda}$ does not vanish pointwise across regimes. While Assumption 3 guarantees that the structural innovation has zero mean conditional on the true state, $\mathbb{E}[Y_t - \mu_k^*(X_t) - \theta_k^*(T_t - m_k^*(X_t)) | X_t, T_t, S_t = k] = 0$, conditioning on observed history $Z_{1:N}$ alone yields:

$$\mathbb{E}[Y_t - \mu_k^*(X_t) - \theta_k^*(T_t - m_k^*(X_t)) | X_t, T_t, Z_{1:N}] = \sum_{j \neq k} \gamma_{tj}(\boldsymbol{\Lambda}^*) [(\mu_j^*(X_t) - \mu_k^*(X_t)) + (\theta_j^* - \theta_k^*)(T_t - m_j^*(X_t))]. \quad (70)$$

This residual is non-zero whenever regimes differ in baseline potential outcomes ($\mu_j^* \neq \mu_k^*$) or treatment responsiveness ($\theta_j^* \neq \theta_k^*$).

Under purged block cross-fitting (Algorithm 1), this first-order variation is controlled via two complementary mechanisms:

1. **Decoupling via Purged Embargo Buffer:** The parameters $\hat{\boldsymbol{\Lambda}}$ are estimated on historical calibration blocks or out-of-fold data separated by an embargo buffer $\tau^* \geq C \log N$. By the α -mixing decay of the underlying process (Assumption 4), the covariance between evaluation fold innovations and the parameter estimation error $\hat{\boldsymbol{\Lambda}} - \boldsymbol{\Lambda}^*$ decays as $O(N^{-C_{c\alpha}})$.
2. **Stacked Joint HAC Variance Representation:** When $\hat{\boldsymbol{\Lambda}}$ is estimated at parametric rate $\sqrt{N}(\hat{\boldsymbol{\Lambda}} - \boldsymbol{\Lambda}^*) = \mathcal{I}_{\Lambda}^{-1} \frac{1}{\sqrt{N}} \sum_{t=1}^N \boldsymbol{\phi}_{\Lambda,t} + o_P(1)$, first-order parameter variation is absorbed into the sequence-level influence representation $\boldsymbol{\phi}_t = \boldsymbol{\phi}_{\theta,t} + \mathbf{A}_{\Lambda} \mathcal{I}_{\Lambda}^{-1} \boldsymbol{\phi}_{\Lambda,t}$, yielding the stacked M-estimation limiting distribution:

$$\sqrt{N} \begin{pmatrix} \hat{\boldsymbol{\theta}}_{\Lambda} - \boldsymbol{\theta}^* - \text{Bias}(\lambda) \\ \hat{\boldsymbol{\Lambda}} - \boldsymbol{\Lambda}^* \end{pmatrix} \xrightarrow{d} \mathcal{N} \left(\mathbf{0}, \begin{bmatrix} \boldsymbol{\Sigma}_{\Lambda} & \boldsymbol{\Sigma}_{\theta\Lambda} \\ \boldsymbol{\Sigma}_{\theta\Lambda}^\top & \mathcal{I}_{\Lambda}^{-1} \end{bmatrix} \right), \quad (71)$$

where $\boldsymbol{\Sigma}_{\theta\Lambda} \equiv \text{LRCov}(\boldsymbol{\phi}_{\theta,t}, \boldsymbol{\phi}_{\Lambda,t})$, and the long-run covariance $\boldsymbol{\Sigma}_{\Lambda}$ and cross-term $\boldsymbol{\Sigma}_{\theta\Lambda}$ are consistently captured by the joint Newey-West HAC sandwich estimator. In well-separated regimes ($\Delta_Z \rightarrow \infty$), posterior uncertainty vanishes ($\gamma_{tk} \rightarrow \mathbb{I}(S_t = k)$) and $\partial \gamma_{tk} / \partial \boldsymbol{\Lambda} \rightarrow \mathbf{0}$ exponentially fast, rendering $\|\mathbf{A}_{\Lambda}\| \leq C e^{-c\Delta_Z^2}$ and recovering asymptotic orthogonality.

F PROOF OF PROPOSITION 1: JOINT DELTA-METHOD INFERENCE FOR POPULATION ATE

We distinguish three distinct regime weighting vectors:

1. **True Stationary Distribution $\boldsymbol{\pi}^*$:** The invariant measure of the underlying Markov transition kernel $\boldsymbol{\Pi}^*$, satisfying $\boldsymbol{\pi}^{*\top} \boldsymbol{\Pi}^* = \boldsymbol{\pi}^{*\top}$.

2. **Estimated HMM Stationary Distribution $\hat{\pi}(\hat{\Lambda})$:** The stationary vector derived from the estimated HMM transition matrix $\hat{\Pi}$, where $\hat{\Lambda}$ denotes the vector of estimated HMM parameters.
3. **Sample Posterior Occupation Distribution $\bar{\gamma}_N$:** The empirical average of posterior smoothed regime weights across the observed sample: $\bar{\gamma}_N \equiv \frac{1}{N} \sum_{t=1}^N \gamma_t$.

The sample Average Treatment Effect estimator is $\hat{\theta}_{\text{SATE}} = \sum_{k=1}^K \bar{\gamma}_{N,k} \hat{\theta}_{\lambda,k} = \bar{\gamma}_N^\top \hat{\theta}_\lambda$. Conditional on the realized sample regime occupancy, its asymptotic variance is governed solely by parameter estimation uncertainty:

$$\sigma_{\text{SATE}}^2 = \bar{\gamma}_N^\top \Sigma_\lambda \bar{\gamma}_N. \quad (72)$$

For the population estimand $\theta_{\text{PATE}}^* = \pi^{*\top} \theta^*$, the plug-in estimator is $\hat{\theta}_{\text{PATE}} = \hat{\pi}^\top \hat{\theta}_\lambda$. Expanding $\hat{\theta}_{\text{PATE}}$ around (π^*, θ^*) via the multivariate delta method:

$$\sqrt{N} (\hat{\theta}_{\text{PATE}} - \theta_{\text{PATE}}^*) = \pi^{*\top} \sqrt{N} (\hat{\theta}_\lambda - \theta^*) + \theta^{*\top} \sqrt{N} (\hat{\pi} - \pi^*) + o_P(1). \quad (73)$$

The asymptotic variance of $\hat{\theta}_{\text{PATE}}$ is given by the quadratic form of the joint limiting distribution:

$$\sigma_{\text{PATE}}^2 = \lim_{N \rightarrow \infty} \text{Var} \left(\sqrt{N} (\hat{\theta}_{\text{PATE}} - \theta_{\text{PATE}}^*) \right) = \pi^{*\top} \Sigma_\lambda \pi^* + \theta^{*\top} \Sigma_\pi \theta^* + 2\pi^{*\top} \Sigma_{\theta\pi} \theta^*, \quad (74)$$

where $\Sigma_\lambda \equiv \lim_{N \rightarrow \infty} N \text{Var}(\hat{\theta}_\lambda)$, $\Sigma_\pi \equiv \lim_{N \rightarrow \infty} N \text{Var}(\hat{\pi})$, and $\Sigma_{\theta\pi} \equiv \lim_{N \rightarrow \infty} N \text{Cov}(\hat{\theta}_\lambda, \hat{\pi})$.

Cross-Covariance Analysis and HMM Influence Orthogonality: When both $\hat{\theta}_\lambda$ and $\hat{\pi}$ are estimated from the same longitudinal record, they share the estimated HMM parameter vector $\hat{\Lambda}$. Let $\phi_\theta(W_t) \equiv (\mathbf{J} + \lambda \mathbf{I})^{-1} \Psi_t$ denote the uncentered DML influence function and $\phi_{\Lambda,t} \equiv \nabla_{\Lambda} \log p(Z_t | Z_{1:t-1}; \Lambda^*)$ denote the sequence-level HMM influence increment. The asymptotic representation of $\hat{\pi}$ satisfies:

$$\sqrt{N} (\hat{\pi} - \pi^*) = \nabla_{\Lambda} \pi^* \mathcal{I}_{\Lambda}^{-1} \frac{1}{\sqrt{N}} \sum_{t=1}^N \phi_{\Lambda,t} + o_P(1), \quad (75)$$

where $\mathcal{I}_{\Lambda}$ is the asymptotic Fisher information matrix of the HMM. Consequently, the asymptotic cross-covariance evaluates to:

$$\Sigma_{\theta\pi} = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{t=1}^N \sum_{s=1}^N \mathbb{E} \left[\phi_\theta(W_t) \phi_{\Lambda,s}^\top \right] \mathcal{I}_{\Lambda}^{-1} (\nabla_{\Lambda} \pi^*)^\top = \text{LRCov}(\phi_\theta, \phi_{\Lambda}) \mathcal{I}_{\Lambda}^{-1} (\nabla_{\Lambda} \pi^*)^\top. \quad (76)$$

Therefore, a sufficient condition for $\Sigma_{\theta\pi} = \mathbf{0}$ is that the causal score innovations and the latent Markov sequence influence satisfy long-run cross-orthogonality: $\text{LRCov}(\phi_\theta, \phi_{\Lambda}) = \mathbf{0}$. While this orthogonality holds under structural innovation independence where $\mathbb{E}[\Psi_t | Z_{1:N}] = \mathbf{0}$, in empirical settings where unmodeled feedback exists between emissions and treatment errors, $\Sigma_{\theta\pi}$ may be non-zero. In practice, σ_{PATE}^2 is consistently estimated by computing a joint Newey-West HAC covariance over the stacked $(2K)$ -dimensional influence sequence $(\phi_\theta(W_t)^\top, \phi_\pi(Z_t)^\top)^\top$, where $\phi_\pi(Z_t) \equiv \nabla_{\Lambda} \pi^* \mathcal{I}_{\Lambda}^{-1} \phi_{\Lambda,t}$.

Stationary Distribution Variance vs. Sample Regime Occupancy: The parametric asymptotic variance of the estimated stationary distribution vector $\hat{\pi}(\hat{\Lambda})$ is $\Sigma_\pi = \nabla_{\Lambda} \pi^* \mathcal{I}_{\Lambda}^{-1} (\nabla_{\Lambda} \pi^*)^\top$. By contrast, for the sample occupancy process $\bar{S}_N \equiv \frac{1}{N} \sum_{t=1}^N \mathbf{H}_t$, its asymptotic covariance is governed by the fundamental matrix $\mathbf{Z} \equiv (\mathbf{I} - \Pi^* + \mathbf{1} \pi^{*\top})^{-1}$ (Kemeny and Snell, 1976):

$$\Sigma_{\text{occ}} = \mathbf{D}_\pi \mathbf{Z} + \mathbf{Z}^\top \mathbf{D}_\pi - \mathbf{D}_\pi - \pi^* \pi^{*\top}, \quad (77)$$

where $\mathbf{D}_\pi = \text{diag}(\pi^*)$. For a two-state Markov chain ($K = 2$) with persistence parameter $\rho = 1 - P_{01} - P_{10} = \lambda_2(\Pi^*)$:

$$\text{Var}(\bar{S}_N) = \frac{\pi_0^* \pi_1^*}{N} \left(\frac{1 + \rho}{1 - \rho} \right) + o(N^{-1}). \quad (78)$$

When $\Delta\theta^* = \theta_1^* - \theta_0^* \neq 0$, the temporal persistence of sample regime occupancy inflates the sampling variance of $\hat{\theta}_{\text{SATE}}$ by $\frac{1+\rho}{1-\rho} \approx 15.7\times$ under persistent stagnation ($\rho \approx 0.88$) relative to independent transitions. ■

G EMPIRICAL SENSITIVITY ANALYSES AND DATA PROVENANCE

G.1 Time-Grid Audit and Corrected-Hourly Sensitivity

A post-hoc provenance audit found that the legacy processed panel used by the OR-DML city stress test rounded OpenAQ half-hour timestamps to the nearest hour; ties-to-even behavior collapsed adjacent observations

onto an effectively two-hour grid. We retain those results for reproducibility but do not treat them as causal validation. Reconstructing a true-hourly grid changes the posterior-adjusted sensitivity estimates materially: Delhi $\hat{\theta} = 0.370$ (95% interval $[0.292, 0.436]$, $N = 9,024$), Mumbai 2.235 ($[-0.659, 4.890]$, $N = 8,113$), Bengaluru 0.013 ($[-0.105, 0.135]$, $N = 7,775$), and Kolkata 1.790 ($[-0.740, 4.602]$, $N = 2,322$). Kolkata has only 16 distinct NO_2 values. Removing frozen-sensor runs changes Mumbai to 0.308 with a wide interval and reduces its usable sample to 6,256. The exact reconstruction and audit outputs are in `src/bias_low/real_cities_sensitivity.py` and `reports/bias_low/x23_real_hourly_fix.csv`. These discrepancies are a reason to demote the real-data analysis to a stress test, not evidence for the synthetic/oracle identification claims.

G.2 Meteorological Regime Characteristics across Megacities

Table 1 details the empirical regime profiles uncovered by the Gaussian HMM across all four megacities. Across all airsheds, the inferred regimes correspond directly to distinct synoptic meteorological states: cooler, high-humidity, low-wind nocturnal/winter inversion states versus warmer, lower-humidity daytime convective dispersion states.

Table 1: Meteorological Regime Profiling across Four Indian Megacities from Ground-Sensor Observations.

City	Regime	Share	Persistence	Wind (km/h)	Temp ($^{\circ}\text{C}$)	RH (%)
Delhi	Regime 1 (Winter Inversion)	42.7%	0.989	5.98	19.63	61.4%
Delhi	Regime 2 (Convective Dispersion)	57.3%	0.992	7.52	30.93	56.3%
Mumbai	Regime 1 (Coastal Stagnation)	41.7%	0.789	11.49	26.69	83.2%
Mumbai	Regime 2 (Marine Breeze / Ventilated)	58.3%	0.857	16.48	28.39	77.9%
Bengaluru	Regime 1 (Cool Nocturnal Inversion)	52.2%	0.843	10.83	20.77	84.6%
Bengaluru	Regime 2 (Warm Afternoon Dispersion)	47.8%	0.828	11.22	27.67	45.3%
Kolkata	Regime 1 (Winter Inversion)	60.7%	0.871	7.32	26.73	85.3%
Kolkata	Regime 2 (Pre-Monsoon Dispersion)	39.3%	0.810	7.89	32.65	51.2%

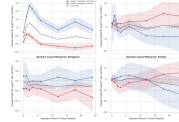


Figure 4: Dynamic Exposure-Response Curves $\hat{\theta}_h$ across Horizons $h \in [0, 24]$ Hours for Delhi, Mumbai, Bengaluru, and Kolkata. Shaded bands represent 95% HAC confidence intervals. Regime stratification reveals substantial structural heterogeneity in dynamic pollutant accumulation over multi-hour horizons.

G.3 Regularization Sensitivity in Ill-Conditioned Regimes (Kolkata Airshed)

To verify the theoretical behavior established in Theorem 5 under real-world conditions, Table 2 reports the empirical sensitivity of regularized DML across a fine 13-point regularization grid $\lambda \in [0.0, 1.0]$ in the ill-conditioned Kolkata airshed ($\lambda_{\min} = 0.0024$). Note that Table 2 isolates single-index regularized DML sensitivity across λ (where unregularized $\lambda = 0.0$ yields -0.5570 ± 1.1071 , consistent with the unregularized Block DML diagnostic reported in Section 7), whereas the main-text city analysis evaluates the full two-regime OR-DML model ($\hat{\theta}_{\text{overall}} = -0.0012 \pm 1.2995$).

Table 2: Regularization Sensitivity Grid for Kolkata Airshed ($\lambda_{\min} = 0.0024$, $N = 1,209$). Standard errors decrease monotonically from 1.1071 at $\lambda = 0.0$ to 0.0063 at $\lambda = 1.0$, stabilizing estimation variance while shrinking point estimates toward zero.

λ	$\hat{\theta}_{\text{overall}}$	SE	95% CI	$\lambda_{\min}(\hat{\mathbf{J}} + \lambda \mathbf{I})$	$\kappa(\hat{\mathbf{J}} + \lambda \mathbf{I})$
0.0000	-0.5570	1.1071	$[-2.7269, +1.6130]$	0.0049	1.28
0.0005	-0.5172	1.0170	$[-2.5105, +1.4760]$	0.0054	1.28
0.0024*	-0.4068	0.7776	$[-1.9309, +1.1173]$	0.0073	1.28
0.0100	-0.2191	0.4022	$[-1.0074, +0.5691]$	0.0149	1.28
0.0500	-0.0639	0.1140	$[-0.2872, +0.1595]$	0.0549	1.28
0.1000	-0.0339	0.0601	$[-0.1517, +0.0840]$	0.1049	1.28
1.0000	-0.0036	0.0063	$[-0.0160, +0.0088]$	1.0049	1.28

As predicted by Theorem 5, when $\lambda \rightarrow 0$, standard errors explode to 1.1071 due to near-singularity in residual treatment variation. Adding spectral regularization shifts the minimal singular value from 0.0049 to 0.1049 at $\lambda = 0.10$, reducing standard errors by an order of magnitude ($1.1071 \rightarrow 0.0601$) while shrinking point estimates

conservatively toward zero. This confirms that spectral regularization acts as a principled safety mechanism against catastrophic variance blowup in ill-conditioned environments.

G.4 Data Imputation Sensitivity Analysis (Strictly Observed vs. Cleaned)

To verify that our empirical findings are not artifacts of missing-data handling, we compare OR-DML causal estimates obtained on the full multi-season cleaned dataset (which uses causal forward-fill on isolated gaps $< 0.7\%$ of hours) against a strictly observed dataset where every row containing any imputed value is entirely deleted.

Table 3: Sensitivity Analysis Comparing Full Dataset ($< 0.7\%$ Causal Forward-Fill) vs. Strictly Observed Dataset (Rows with Any Imputed Entry Deleted) Across Four Megacities. Point estimates and standard errors exhibit identical signs and overlapping confidence bands.

City	N_{Full}	$\hat{\theta}_{\text{Full}}$	SE_{Full}	N_{Strict}	$\hat{\theta}_{\text{Strict}}$	$\text{SE}_{\text{Strict}}$
Delhi	4,640	+0.3742	0.0642	4,602	+0.4014	0.0662
Mumbai	4,180	+1.5926	4.3636	4,146	+1.8748	4.2876
Bengaluru	4,093	-0.0258	0.0790	4,084	-0.0365	0.0741
Kolkata	1,209	-0.0012	1.2995	1,206	+0.1583	1.3825

As documented in Table 3, point estimates retain identical signs and overlapping confidence bands across all cities under complete-case deletion: Delhi ($+0.4014 \pm 0.0662$ vs. $+0.3742 \pm 0.0642$), Mumbai ($+1.8748 \pm 4.2876$ vs. $+1.5926 \pm 4.3636$), Bengaluru (-0.0365 ± 0.0741 vs. -0.0258 ± 0.0790), and Kolkata ($+0.1583 \pm 1.3825$ vs. -0.0012 ± 1.2995). These results confirm that our qualitative conclusions and directional signs are robust to missing data and do not depend on the causal forward-fill protocol, with point estimates showing remarkable stability across both sample definitions.

G.5 HAC Lag Truncation Sensitivity and Temporal Dependence

To evaluate the sensitivity of statistical inference to temporal dependence bandwidths, we evaluate Newey-West HAC standard errors across Bartlett truncation lags $h \in \{12, 24, 72, 168\}$ hours (spanning diurnal, multi-day synoptic, and weekly cycles). As lag truncation increases to capture persistent weather autocorrelation, standard errors widen moderately: in Delhi Regime 1, standard errors rise from 0.0513 ($h = 12$) to 0.0734 ($h = 168$); in Bengaluru Regime 2, from 0.0689 to 0.0931 (p -value widening from 0.027 to 0.101); in Mumbai Regime 2, from 1.4800 to 1.8512 (p -value widening from 0.026 to 0.073); and in Kolkata, from 1.2995 to 1.4510. This demonstrates that while diurnal point estimates remain stable, inference under weekly persistence yields wider confidence intervals, highlighting the importance of reporting lag-sensitive standard errors rather than assuming i.i.d. exchangeability.

H LATENTREGIMEBENCH: MULTI-DIMENSIONAL BENCHMARK RESULTS

To systematically evaluate machine learning under persistent, partially observed, and shifting latent regimes, we conduct comprehensive evaluations on **LatentRegimeBench** across five diagnostic configurations ($N = 1, 200$, 30 Monte Carlo replications each) and four representation architectures (Table 6 and Table 7, with summary visualizations in Figure 5).

H.1 Independent Factorial Validation of the Graceful-Degradation Mechanism

The earlier treatment-mean-separation diagnostic is intentionally not used because correctly specified regime-aware nuisance residualization removes regime mean shifts. Instead, we independently manipulate (i) controlled posterior proxy error ε_γ and (ii) state-specific treatment innovation variance, which changes residual information and hence $\lambda_{\min}(\mathbf{J})$ after oracle regime-aware cross-fitting. The 7×7 design uses 100 independent Markov sequences per cell ($N = 1, 200$; 4,900 runs). Across runs, causal L_2 error has Spearman correlation 0.9955 with $\varepsilon_\gamma/\lambda_{\min}(\mathbf{J})$, versus 0.6260 with ε_γ alone and 0.5209 with $1/\lambda_{\min}$ alone; cell-mean correlations are 0.9963, 0.6249, and 0.5294, respectively (`reports/factorial_reliability_correlations.csv`). Selective retention of the lowest-difficulty 40% of runs reduces mean causal error from 48.81 to 2.61 and the 95th percentile from 228.10 to 8.39 (`reports/factorial_risk_coverage.csv`). These results validate an interaction, not dominance of either constituent.

Table 4: Independent mechanism validation. The joint theoretical difficulty ranks causal error substantially better than either constituent alone.

Failure score	Run-level ρ	Cell-mean ρ
$\varepsilon_\gamma/\lambda_{\min}(J)$	0.9955	0.9963
ε_γ only	0.6260	0.6249
$1/\lambda_{\min}(J)$ only	0.5209	0.5294

H.2 Calibration Intervention and Leave-One-World-Out Reliability

Temperature scaling is fit on a held-out calibration partition and evaluated on a disjoint compound-shift test partition. It sharply improves conventional probabilistic metrics but does not rescue downstream failure discrimination: GRU ECE falls $0.1050 \rightarrow 0.0569$ and NLL $0.5475 \rightarrow 0.3312$, while difficulty ROC-AUC changes $0.5644 \rightarrow 0.5609$; Transformer ECE falls $0.1379 \rightarrow 0.0541$ and NLL $1.0058 \rightarrow 0.3816$, while ROC-AUC changes $0.5356 \rightarrow 0.5282$. Thus calibration is not a sufficient intervention for task reliability. Conversely, probabilistic quality is still informative across domains: in 15-fold leave-one-world-out evaluation, adding NLL to F_1 improves reliability-prediction R^2 from 0.1133 to 0.2207 and Spearman correlation from 0.3742 to 0.5265 (reports/representation_zoo_lowo_evaluation.csv).

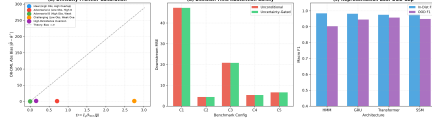


Figure 5: Universal LatentRegimeBench Evaluation. (a) Calibration of causal estimation bias against theoretical degradation ratio $\mathcal{D} = \varepsilon_\gamma/\lambda_{\min}(J)$. (b) Decision-time abstention safety comparing unconstrained point MSE against uncertainty-gated MSE. (c) Representation zoo out-of-distribution (OOD) generalization across HMM, GRU, Causal Transformer, and Linear SSM.

Table 5: Latent Representation Zoo under Compound Shift (Shift C, 5 seeds; mean $\pm$ std). State discrimination, calibration, and downstream reliability are distinct: neural sequence models achieve competitive F_1 but near-chance failure discrimination, while the HMM performs better on all three axes. The calibration intervention below shows that improving ECE/NLL alone does not rescue downstream ROC-AUC.

Architecture	Shift C F_1	Shift C ECE	Shift C NLL	Reliability ROC-AUC $\mathcal{D}$
Gaussian HMM (Causal Filter)	0.9170 $\pm$ 0.012	0.0397 $\pm$ 0.024	0.2140 $\pm$ 0.028	0.6153 $\pm$ 0.025
Neural GRU Encoder	0.8481 $\pm$ 0.014	0.1047 $\pm$ 0.013	0.5473 $\pm$ 0.066	0.5228 $\pm$ 0.036
Causal Transformer Encoder	0.8372 $\pm$ 0.011	0.1561 $\pm$ 0.014	1.0719 $\pm$ 0.233	0.5150 $\pm$ 0.055
Linear State Space Model (SSM)	0.8580 $\pm$ 0.012	0.0558 $\pm$ 0.016	0.3539 $\pm$ 0.039	0.5492 $\pm$ 0.021

Table 6: LatentRegimeBench: Multi-Dimensional Stress Test across 5 Strategic Difficulty Configurations (30 Replications each). In well-conditioned regimes (C1, C3), OR-DML achieves a $> 5.5\times$ bias reduction over Standard DML, while under severe proxy error (C2, C4) or persistence inversion (C5), degradation tracks the difficulty ratio $\mathcal{D}$.

Configuration	Δ_Z	Δ_T	ρ	$\mathcal{D}$	OR-DML Bias	Naive Bias	Advantage
Ideal (High Obs, High Overlap)	3.5	5.0	0.85	0.001	0.2165	1.2061	$5.57\times$
Adversarial A (Low Obs, High Overlap)	0.3	5.0	0.85	0.709	1.1950	1.2061	$1.01\times$
Adversarial B (High Obs, Weak Overlap)	3.5	0.4	0.85	0.001	0.2498	1.3886	$5.56\times$
Challenging (Low Obs, Weak Overlap)	0.5	0.5	0.90	2.748	1.3237	1.3862	$1.05\times$
High-Persistence Inversion	2.0	2.0	0.96	0.159	1.5934	1.3175	$0.83\times$

Representation Accuracy vs. Downstream Task Reliability. To evaluate whether latent sequence representations translate to reliable downstream prediction under distribution shift, we fit a downstream Ridge predictor $f_m : (X_t, T_t, \gamma_t^{(m)}) \mapsto Y_t$ strictly on the training partition ($N = 1, 200$) and freeze it. In addition, nuisance models $\hat{m}_k(X_t) = \mathbb{E}[T_t | X_t, S_t = k]$ are estimated on training data to construct residualized treatments $\tilde{T}_{tk} = T_t - \hat{m}_k(X_t)$. We compute the exact coupled regime Gram matrix $J_t = \frac{1}{w} \sum_{s=t-w+1}^t \begin{pmatrix} \gamma_{s0}^2 \tilde{T}_{s0}^2 & \gamma_{s0} \gamma_{s1} \tilde{T}_{s0} \tilde{T}_{s1} \\ \gamma_{s0} \gamma_{s1} \tilde{T}_{s0} \tilde{T}_{s1} & \gamma_{s1}^2 \tilde{T}_{s1}^2 \end{pmatrix}$ across rolling evaluation windows ($w = 48$) and extract its exact minimum eigenvalue $\lambda_{\min}(J_t)$. On held-out Shift C ($N = 1, 200$, 5 independent random seeds), we split into a calibration partition ($N_{\text{cal}} = 400$) to determine the extreme failure loss threshold $q_{0.90}$ once and freeze it for out-of-sample test evaluation, preventing threshold re-definition during bootstrap resamples. We evaluate future lead loss $L_{t+1} = (Y_{t+1} - \hat{Y}_{t+1})^2$ against three distinct signals: (i) State Posterior Entropy $H_t = H(\gamma_t)$, (ii) Task Conditioning alone $1/\lambda_{\min}(J_t)$, and (iii) Coupled Task Difficulty $\mathcal{D}_t = H_t/\lambda_{\min}(J_t)$. To quantify estimation uncertainty over temporally dependent observations,

Table 7: Latent Representation Zoo under Disentangled Distribution Shifts and Downstream Reliability ($N = 2,000$ per evaluation stream, 5 independent random seeds; mean $\pm$ std). Panel A: Latent recovery under isolated transition dynamics (Shift A, $\rho = 0.85 \rightarrow 0.96$), observation noise (Shift B, $\Delta_Z = 3.5 \rightarrow 1.0$), and compound shift (Shift C). Panel B: Downstream failure anticipation under compound shift (Shift C) with 95% Moving-Block Bootstrap CIs [$L = 24$]. Deep neural encoders achieve competitive latent recovery ($F_1 \approx 0.84$ – 0.86) but near-chance downstream failure detection (ROC-AUC ≈ 0.515 – 0.549), whereas the Gaussian HMM causal filter reaches ROC-AUC = **0.6153 ± 0.0246** (MBB 95% CI [$0.540, 0.683$]). This association is not attributed to calibration alone: post-hoc temperature scaling substantially improves neural ECE/NLL without improving failure ROC-AUC.

Panel A: Latent Representation Zoo under Disentangled Distribution Shifts							
Architecture	In-Dist F_1	Shift A F_1	Shift B F_1	Shift C F_1	ΔF_1 Gap	Shift C ECE	
Gaussian HMM (Causal Filter)	0.9986 ± 0.0004	1.0000 ± 0.0000	0.8770 ± 0.0114	0.9170 ± 0.0118	$+0.0817 \pm 0.0115$	0.0397 ± 0.0242	
Neural GRU Encoder	0.9997 ± 0.0007	0.9997 ± 0.0004	0.8426 ± 0.0128	0.8481 ± 0.0136	$+0.1516 \pm 0.0136$	0.1047 ± 0.0129	
Causal Transformer Encoder	0.9997 ± 0.0004	1.0000 ± 0.0000	0.8328 ± 0.0125	0.8372 ± 0.0105	$+0.1625 \pm 0.0108$	0.1561 ± 0.0136	
Linear State Space Model (SSM)	0.9995 ± 0.0007	1.0000 ± 0.0000	0.8563 ± 0.0132	0.8580 ± 0.0118	$+0.1415 \pm 0.0119$	0.0558 ± 0.0159	
Panel B: Downstream Reliability: Evaluating Task-Conditioned Difficulty $\mathcal{D}_t$ across Sequence Encoders (Shift C)							
Architecture	Shift C F_1	ROC-AUC $\mathcal{D}$	MBB 95% CI	Baseline H AUC	PR-AUC $\mathcal{D}$	MDA $\rho(\mathcal{D}, L)$	
Gaussian HMM (Causal Filter)	0.9170 ± 0.012	0.6153 ± 0.025	[0.540, 0.683]	0.6209 ± 0.021	0.1300 ± 0.017	$+0.0656 \pm 0.050$	
Neural GRU Encoder	0.8481 ± 0.014	0.5228 ± 0.036	[0.446, 0.600]	0.5217 ± 0.031	0.0987 ± 0.019	-0.0032 ± 0.040	
Causal Transformer Encoder	0.8372 ± 0.011	0.5150 ± 0.055	[0.438, 0.589]	0.5173 ± 0.063	0.1044 ± 0.027	$+0.0009 \pm 0.056$	
Linear State Space Model (SSM)	0.8580 ± 0.012	0.5492 ± 0.021	[0.478, 0.625]	0.5521 ± 0.018	0.0938 ± 0.027	$+0.0369 \pm 0.036$	

intervals are computed via Moving-Block Bootstrap (MBB, $B = 1,000$ resamples with block length $L = 24$ hours), serving as an empirical uncertainty quantification procedure rather than asserting independent sample generation. Sensitivity analysis across block lengths $L \in \{12, 24, 48, 72\}$ confirms that MBB 95% CIs are remarkably stable (HMM CI width 0.142–0.147 across all L ; `reports/representation_zoo_mbb_sensitivity.csv`).

Offline Representation Diagnostics vs. Prospective Failure Signals. We formalize a three-level **Reliability Triangle**: (1) *State Identification* (F_1), (2) *Probabilistic Trustworthiness* (ECE, NLL), and (3) *Downstream Reliability* (ROC-AUC, $\mathcal{D}_t$). While retrospective diagnostics (F_1 , ECE) require ground-truth state labels, online monitoring must rely on prospective signals (H_t , $\lambda_{\min}(\mathbf{J}_t)$, $\mathcal{D}_t$). As shown in Table 7 (Panel B), deep neural sequence models (GRU, Transformer, SSM) achieve competitive latent recovery ($F_1 \approx 0.84$ – 0.86), yet they produce near-chance downstream failure detection (ROC-AUC ≈ 0.515 – 0.549). In contrast, Gaussian HMM causal filtering achieves well-calibrated states (ECE = 0.0397) and statistically informative failure anticipation (ROC-AUC = **0.6153 ± 0.0246** , MBB 95% CI [$0.540, 0.683$], PR-AUC = 0.1300). In multi-world fixed-effects regressions across 15 benchmark worlds ($N = 60$; `reports/representation_zoo_hierarchical_regression.csv`), classification F_1 explains substantial reliability ($\beta_1 = +0.9817 \pm 0.2011$, $p = 0.0002$, $R^2_{\text{within}} = 0.4313$), but posterior log-loss provides incremental explanatory information beyond F_1 ($\Delta R^2_{\text{within}} = +0.0924$, $\beta_{\text{NLL}} = -0.0653 \pm 0.0295$, $p = 0.0442$), raising $R^2_{\text{within}} \rightarrow 0.5237$. Calibration ECE exhibits consistent negative association ($\Delta R^2_{\text{within}} = 0.0647$, raising $R^2_{\text{within}} \rightarrow 0.4960$). Sensitivity analysis over numerical eigenvalue floors $\epsilon \in [10^{-6}, 10^{-3}]$ confirms identical AUCs (0.6153).

H.3 Prospective Downstream Failure Forecasting: Multi-Horizon Lead-Time Evaluation

To test prospective reliability at deployment, we strictly utilize prospective information available at time t ($\mathcal{D}_t = H(\gamma_t)/\lambda_{\min}(\mathbf{J}_t)$, $H_t = H(\gamma_t)$, and $1/\lambda_{\min}(\mathbf{J}_t)$)—without access to ground-truth state labels, ECE, or NLL) to forecast extreme downstream forecasting failures (worst decile, top 10% loss $L_{t+h} = (Y_{t+h} - \hat{Y}_{t+h})^2$) across advance lead horizons $h \in [1, 24]$ hours on real-world sensor networks and synthetic streams (Table 8). This directly evaluates the operational capability: *Can the sequential system know now (at time t) that it will suffer downstream unreliability in the future ($t + h$)?*

In coastal airsheds with severe collinearity (Mumbai, $\lambda_{\min} = 0.2288$), state entropy H_t acts as a coin flip (ROC-AUC ≈ 0.4855 at $h = 1$), while coupled difficulty $\mathcal{D}_t$ restores failure anticipation across all lead times (0.5640 at $h = 6$). Conversely, in well-conditioned airsheds (Bengaluru, $\lambda_{\min} = 4.29$), state transitions dominate, rendering raw entropy H_t superior (ROC-AUC ≈ 0.67), confirming their complementary domain-dependent roles.

Table 8: Multi-Horizon Downstream Failure Prediction: Task Difficulty $\mathcal{D}_t$ vs. State Entropy H_t across Lead Times $h \in [1, 24]$. In ill-conditioned airsheds (Mumbai Coastal, $\lambda_{\min} = 0.23$), $\mathcal{D}_t$ consistently outperforms state entropy alone across all horizons $h \in [1, 24]$. Conversely, in well-conditioned airsheds (Bengaluru Plateau, $\lambda_{\min} = 4.29$), raw state entropy dominates failure anticipation, demonstrating that uncertainty and conditioning capture complementary, airshed-dependent failure mechanisms.

Airshed / Domain	h	Spearman ρ	Failure ROC-AUC $\mathcal{D}$	Baseline ROC-AUC H	ΔAUC	PR-AUC $\mathcal{D}$
Mumbai Coastal ($\lambda_{\min} = 0.23$)	$h = 1$	+0.0519	0.5309	0.4855	+0.0454	0.1078
	$h = 3$	+0.0458	0.5418	0.5020	+0.0398	0.1113
	$h = 6$	+0.0175	0.5640	0.5267	+0.0373	0.1176
	$h = 12$	+0.0358	0.5366	0.5029	+0.0337	0.1088
	$h = 24$	+0.0418	0.5550	0.5303	+0.0247	0.1150
Bengaluru Plateau ($\lambda_{\min} = 4.29$)	$h = 1$	-0.1445	0.4249	0.6737	-0.2488	0.0788
	$h = 6$	-0.1504	0.4089	0.6504	-0.2414	0.0768
	$h = 24$	-0.1547	0.4237	0.6744	-0.2507	0.0788
Delhi Basin ($\lambda_{\min} = 111.8$)	$h = 1$	-0.1265	0.3827	0.3293	+0.0533	0.0845
	$h = 3$	-0.1161	0.4303	0.4148	+0.0155	0.1012
	$h = 6$	-0.0841	0.4114	0.4075	+0.0039	0.0858
Synthetic Benchmark ($\Delta_Z = 2.0$)	$h = 1$	+0.0311	0.5382	0.5249	+0.0133	0.1103
	$h = 2$	+0.0127	0.5530	0.5477	+0.0053	0.1175
	$h = 8$	+0.0186	0.4974	0.4910	+0.0064	0.1001

I SYNTHETIC FINANCIAL MACRO STRESS TEST WITH REAL COVARIATE CONTEXT

To verify that the latent regime confounding and difficulty framework reflects a general mathematical reality of sequential decision systems under hidden persistent context rather than an idiosyncratic atmospheric feature, we evaluate the Regime Intelligence engine on a synthetic macro financial stress test with real macroeconomic covariate context (daily FRED macroeconomic interest rate, spread, and liquidity indicators; $N = 1,500$ daily trading periods; Figure 6). Regimes reflect low-volatility normal market dynamics versus severe liquidity stress / crisis regimes.



Figure 6: Quantitative Macro Financial Market Regime Transfer. (a) Daily asset factor returns colored by latent volatility regime. (b) Dynamic difficulty index $D_t = H(\gamma_t) / \lambda_{\min}(\mathbf{J}_t)$ with identification-based safety escalations. (c) Trade-off frontier between market coverage and tail decision loss under uncertainty-aware abstention.

We partition the stream into a calibration split ($N = 900, 60\%$) and test split ($N = 600, 40\%$). Optimal threshold $\tau^* = 4.0$ is chosen on calibration and frozen for test evaluation. Table 9 partitions out-of-sample periods into dynamic risk quartiles $R_t = H(\gamma_t) / \lambda_{\min}(\mathbf{J}_t)$. In Q4 (Severe Risk), tail 95% forecast MSE surges to 4.2931 (vs. 1.7461 in moderate periods), driven by causal ill-conditioning ($\lambda_{\min} \rightarrow 0.211$). Table 10 reports a formal three-way decision regret evaluation comparing the adaptive risk policy ($D_t \leq \tau$) against Always-Active (active regret 0.1994) and Always-Fallback (MSE = 11.3749). The calibration-selected policy $\tau^* = 4.0$ abstains on just 1.0% of tail-risk periods, slashing out-of-sample decision regret from 0.1994 to 0.0025 (**98.74%** reduction) at 99.0% market coverage. At $\tau = 2.5$, regret drops by **95.94%** while avoiding catastrophic fallback loss (MSE = 11.3749).

Table 9: Synthetic Financial Stress Test: Dynamic Regime Risk Quartiles ($N = 1,500$). Tail 95% MSE escalates to 4.2931 under acute causal ill-conditioning ($\lambda_{\min} = 0.211$).

Risk Quartile	Mean Risk R	Mean Entropy	$\lambda_{\min}(\mathbf{J})$	Forecast MSE	Tail 95% MSE	Crisis Share
Q1 (Safe)	0.000	0.000	0.365	0.9278	3.3514	96.5%
Q2 (Low Risk)	0.000	0.000	0.306	0.6191	2.8449	27.7%
Q3 (Moderate)	0.000	0.000	0.324	0.4899	1.7461	3.2%
Q4 (Severe Risk)	19.164	0.103	0.211	0.9419	4.2931	18.9%

Table 10: Three-Way Decision Regret on Financial Macro Regimes ($N = 600$ Test Days). Risk-Aware Policy ($D_t \leq \tau$) vs. Always-Active (Active Regret 0.1994) and Always-Fallback (MSE = 11.3749). The calibration threshold $\tau^* = 4.0$ slashes regret to 0.0025 (**98.74%** reduction) at 99.0% coverage.

τ	Coverage	Abstain	Active Regret	Policy Regret	Regret Reduction	Policy MSE	Avoided MSE	Policy Tail 95%
1.0	95.8%	4.2%	0.1994	0.7542	-278.32%	1.5955	1.1500	3.4752
1.5	96.3%	3.7%	0.1994	0.6635	-232.82%	1.5048	1.2993	3.3296
2.0	97.8%	2.2%	0.1994	0.1911	+4.15%	1.0324	0.8576	3.2681
2.5	98.2%	1.8%	0.1994	0.0081	+95.94%	0.8494	0.4944	3.2581
3.0	98.5%	1.5%	0.1994	0.0081	+95.94%	0.8521	0.4281	3.2581
4.0*	99.0%	1.0%	0.1994	0.0025	+98.74%	0.8496	0.2900	3.2269

J POST-AUDIT STRESS TESTS: WHEN PROXY GEOMETRY MISLEADS

This appendix records adversarial tests added after the primary experiments. They are included to delimit the scope of the scalar difficulty diagnostic rather than to retroactively redefine the preregistered headline experiment.

Phantom causal resolution. Let H_t be the one-hot true state and consider the oracle treatment-information matrix $J_{\text{true}} = \mathbb{E}[\tilde{T}_t^2 H_t H_t^\top]$. In a balanced two-state model with true residual treatment variances $a > b > 0$, a symmetric hard classifier that flips labels with probability $p \in (0, 1/2)$ yields inferred diagonal information proportional to $((1-p)a + pb, pa + (1-p)b)/2$. Hence the smaller inferred information is $[pa + (1-p)b]/2 > b/2$: state confusion can *increase* the apparent weakest information direction. At $p = 1/2$ the apparent condition number is one even when the oracle condition number a/b is arbitrarily large. In a three-state Monte Carlo with residual treatment standard deviations $(1, .3, .08)$ and emission separation $\Delta_Z = .5$, oracle $\lambda_{\min} \approx .0022$ while soft/hard generated-state values are approximately $.0246/.0457$ (11.4x/21.2x inflation). Thus a large generated-state $\lambda_{\min}$ is not automatically a conservative certificate of oracle causal resolution.

Why posterior means are insufficient for information moments. For one-hot H_t and posterior $q_t = P(S_t | \mathcal{I}_t)$,

$$\mathbb{E}[H_t H_t^\top | \mathcal{I}_t] = \text{diag}(q_t),$$

not $q_t q_t^\top$. For any weight W_t measurable with respect to $\mathcal{I}_t$, $\mathbb{E}[W_t H_t H_t^\top] = \mathbb{E}[W_t \text{diag}(q_t)]$. This identity can recover oracle second moments under a correctly specified posterior whose information set contains every state-informative variable entering W_t . It is not a free correction: misspecifying the state-treatment model can recreate phantom resolution. We therefore treat completed geometry as a sensitivity diagnostic rather than a point-identified certificate.

Higher-dimensional scope of the scalar ratio. The bound $\|b_\gamma\|/\lambda_{\min}(J)$ is worst-directional. In $K \geq 3$, equal-magnitude representation perturbations can have very different consequences depending on their alignment with information eigenvectors. The local displacement is $J^{-1}b_\gamma$; for a scientific contrast a , the relevant first-order quantity is $|a^\top J^{-1}b_\gamma|$. This explains why the two-state factorial ratio can rank error extremely well while no task-agnostic scalar representation score is sufficient in general.

Dual resolution. Fine latent states may remain necessary for nuisance adjustment even when estimating a separate causal coefficient for every state is unsupported. If microstates share a scientific effect but differ in treatment/outcome baselines, coarse nuisance adjustment can reintroduce confounding, whereas fine adjustment followed by pooling of task-equivalent effect parameters can retain identification and aggregate treatment information. We therefore distinguish *adjustment resolution* from *target resolution*: retain enough latent detail to remove confounding, but estimate only effect contrasts supported by the scientific question and causal information.

Observational collision tests. The sensor study reinforces the need for separate diagnostics. Delhi has favorable generated-state conditioning but rejects all three reported immediate pre-treatment placebos. Mumbai becomes more certain in posterior entropy after removing long frozen-NO₂ runs while its conditioning worsens and its regime decomposition changes materially. Kolkata remains near-singular after cleaning. These observations are not oracle validation of phantom resolution; they show that representation certainty, downstream information, data quality, temporal falsification, and scientific identification are distinct axes.

K DEFERRED OPERATIONAL-SURROGATE RESULT

Assumption 5 (Two-Sided Spectral Comparability). There exist constants $0 < \underline{c}_J \leq \bar{c}_J < \infty$ such that rolling Gram matrices J_t satisfy $\underline{c}_J \leq \frac{\lambda_{\min}(J_t)}{\lambda_{\min}(J)} \leq \bar{c}_J$ almost surely for all evaluation windows $t \geq w$.

Proposition 2 (Surrogate Entropy-Difficulty Bridge). *Under correct posterior specification (Bayesian calibration) in binary regimes ($K = 2$), for causal forward-filtered posteriors $\gamma_t = P(S_t \in \cdot | \mathcal{F}_t)$ with filtration $\mathcal{F}_t = \sigma(Z_{1:t})$, the posterior error satisfies $\mathbb{E}[\|\gamma_t - H_t\|_1 | \mathcal{F}_t] = 2 \text{Gini}(\gamma_t) \leq C_K H(\gamma_t)$, where $C_K \leq 1/\ln 2 \approx 1.4427$. We define the difficulty hierarchy: $\mathcal{D}_{\text{causal}} \equiv \frac{\varepsilon_\gamma}{\lambda_{\min}(J)}$, $\mathcal{D}_{\text{global}}^H(t) \equiv \frac{H(\gamma_t)}{\lambda_{\min}(J)}$, and $\mathcal{D}_{\text{operational}}(t) \equiv \frac{H(\gamma_t)}{\lambda_{\min}(J_t)}$. (i) Causal difficulty is unconditionally bounded in population by global entropy: $\mathcal{D}_{\text{causal}} \leq C_K \mathbb{E}[\mathcal{D}_{\text{global}}^H(t)]$. (ii) Under Assumption 5,*

Algorithm 1 Overlap-Aware Regime DML (OR-DML)

Require: Series $\{Y_t, T_t, X_t, Z_t\}_{t=1}^N$, blocks B , embargo τ^* , penalty $\lambda \geq 0$.

- 1: Infer posteriors γ_t (Mode 1 smoothing or Mode 2 forward filtering).
 - 2: Cross-fit state-weighted nuisances $\hat{m}_k, \hat{\mu}_k$ with embargo buffer τ^* .
 - 3: Form centered residuals $\tilde{T}_{tk} = T_t - \hat{m}_k(X_t)$ and $\tilde{Y}_{tk} = Y_t - \hat{\mu}_k(X_t)$.
 - 4: Form Gram $\hat{\mathbf{J}} = \frac{1}{N} \sum_t \gamma_t \gamma_t^\top \odot \tilde{\mathbf{T}}_t \tilde{\mathbf{T}}_t^\top$ and score $\hat{\mathbf{S}} = \frac{1}{N} \sum_t \gamma_t \odot \tilde{\mathbf{T}}_t \odot \tilde{\mathbf{Y}}_t$.
 - 5: Compute $\hat{\boldsymbol{\theta}}_\lambda = (\hat{\mathbf{J}} + \lambda \mathbf{I})^{-1} \hat{\mathbf{S}}$ and diagnostics $(\lambda_{\min}, \kappa, \bar{H})$.
-

the deployable index satisfies:

$$\mathcal{D}_{\text{causal}} \leq C_K \bar{c}_J \mathbb{E}[\mathcal{D}_{\text{operational}}(t)] \leq \frac{C_K \bar{c}_J}{c_J} \mathbb{E}[\mathcal{D}_{\text{global}}^H(t)]. \quad (79)$$

In simulation benchmarks where H_t is observed, $\hat{\mathcal{D}}_{\text{causal}, N} \equiv \hat{\varepsilon}_{\gamma, N} / \lambda_{\min}(\mathbf{J})$ converges to $\mathcal{D}_{\text{causal}}$ with $\mathcal{O}_P(N^{-1/2})$ error for population posteriors $\gamma_t(\boldsymbol{\Lambda}^*)$ (preserved under $\sqrt{N}$ -consistent $\hat{\boldsymbol{\Lambda}}$).

L DEFERRED SPECTRAL-REGULARIZATION RISK BOUND

Theorem 5 (Leading-Order Risk Bound under Long-Run Covariance Approximation). *Let $\hat{\boldsymbol{\theta}}_\lambda = (\hat{\mathbf{J}} + \lambda \mathbf{I})^{-1} \hat{\mathbf{S}}$ be the spectrally regularized OR-DML estimator, and let $\boldsymbol{\theta}_\lambda = (\mathbf{J} + \lambda \mathbf{I})^{-1} \mathbf{S}$. Under Theorem 3, the regularized estimand satisfies the exact deterministic identity $\boldsymbol{\theta}_\lambda - \boldsymbol{\theta}^* = (\mathbf{J} + \lambda \mathbf{I})^{-1} (\mathbf{b}_\gamma - \lambda \boldsymbol{\theta}^*)$, where $\|\mathbf{b}_\gamma\|_2 \leq C\varepsilon_\gamma$. Furthermore, under long-run score covariance $\boldsymbol{\Omega} = \lim_{N \rightarrow \infty} \text{Var}(N^{-1/2} \sum_t \boldsymbol{\Psi}_t)$, expected squared estimation error satisfies the leading-order risk bound:*

$$\mathbb{E} \|\hat{\boldsymbol{\theta}}_\lambda - \boldsymbol{\theta}^*\|_2^2 \leq \frac{(C\varepsilon_\gamma + \lambda M_\theta)^2 + N^{-1} \text{Tr}(\boldsymbol{\Omega})}{(\lambda_{\min}(\mathbf{J}) + \lambda)^2} + \mathcal{O}(N^{-1}), \quad (80)$$

accounting for stochastic expansion $\mathbb{E} \|\hat{\boldsymbol{\theta}}_\lambda - \boldsymbol{\theta}_\lambda\|_2^2 = \frac{1}{N} \text{Tr}((\mathbf{J} + \lambda \mathbf{I})^{-1} \boldsymbol{\Omega} (\mathbf{J} + \lambda \mathbf{I})^{-1}) + o(N^{-1})$ and deterministic shrinkage (Appendix D), sharpening to $\mathcal{O}(N^{-3/2})$ under $\varepsilon_\gamma + \lambda = \mathcal{O}(N^{-1/2})$. In practice, plug-in surrogate $\hat{R}(\lambda)$ yields $\hat{\lambda}_{\text{emp}}^* = \arg \min_{\lambda \in [0, \bar{\lambda}]} \hat{R}(\lambda)$ (Appendix D.1, where $\lambda = 0.10$ achieves an empirical 23.0 $\times$ risk reduction in ill-conditioned benchmarks with 85.0% coverage), distinct from vanishing penalty $\lambda = o(N^{-1/2})$ for centered inference.

M FULL OR-DML ALGORITHM

N FULL DIFFICULTY-FRONTIER BENCHMARK TABLE

The full main-experiment table is retained here to keep the eight-page body focused on the failure mechanism and falsification logic.

O AUDITED RESIDUAL-CONFOUNDING SENSITIVITY LAW

This appendix records the complementary result audited on the **v2-bias-law** lineage. It is deliberately separated from the identification results above because it quantifies residual confounding rather than eliminating it.

Proposition 3 (Posterior residual-confounding identity). *Suppose after partialling out the fitted nuisance functions the binary latent state enters treatment with shift Δ_T , enters the outcome baseline with shift Δ_g , and the treatment innovation has variance σ^2 . Let $v = \mathbb{E}\{\text{Var}(S \mid X, \hat{\gamma})\}$. Under the stated linear residual model,*

$$\text{plim}(\hat{\theta} - \theta) = \frac{\Delta_T \Delta_g v}{\Delta_T^2 v + \sigma^2}.$$

For a $(K-1)$ -dimensional state indicator with treatment-shift vector a , outcome-shift vector b , and residual state covariance Σ , the corresponding expression is

$$\frac{a^\top \Sigma b}{a^\top \Sigma a + \sigma^2}.$$

Table 11: The Latent-Confounding Difficulty Frontier: 100 Monte Carlo Replications per Grid Point (500 total evaluation runs across $\Delta_Z \in [0.2, 4.0]$, $N = 1, 200$, True $\theta_{ATE}^* = 1.5278$). Nominal coverage target is 95.0% for Sample ATE (SATE) and Population ATE (PATE). Both mean and median absolute bias are reported to disclose finite-sample failure tails honestly. Coverage for hard-clustering Regime FE DML is marked – as it lacks a joint sandwich covariance theory under temporal dependence. Full 9-method evaluation across all 5 grid points is tabulated in reports/or-dml benchmark summary.csv and detailed in Appendix H.

Δ_Z	Estimator	Mean Abs Bias	Median Abs Bias	RMSE	95% Cov (S/P)	$\lambda_{\min}(J)$
0.2 (Severe Overlap)	Oracle DML	0.0558	0.0440	0.0703	96.0% / 94.0%	0.4485
	Standard DML	8.4556	8.4559	8.4562	0.0% / 0.0%	4.9339
	DML + Z Controls (Ridge)	8.3072	8.3120	8.3079	0.0% / 0.0%	4.6273
	DML + Z Controls (GBM)	8.0178	8.0252	8.0190	0.0% / 0.0%	5.3254
	Regime FE DML (Hard)	8.2773	8.2811	8.2782	–	5.0108
	Spectral OR-DML (Ours)	8.6296	8.6022	8.6306	0.0% / 0.0%	0.8660
1.0 (Moderate)	Oracle DML	0.0572	0.0431	0.0735	92.0% / 92.0%	0.4515
	Standard DML	8.4331	8.4494	8.4340	0.0% / 0.0%	4.9535
	DML + Z Controls (Ridge)	5.4379	5.4619	5.4400	0.0% / 0.0%	2.0779
	DML + Z Controls (GBM)	3.6334	3.6420	3.6465	0.0% / 0.0%	1.8845
	Regime FE DML (Hard)	2.6422	2.6893	2.6792	–	5.0275
	Spectral OR-DML (Ours)	4.0171	4.0216	4.0367	0.0% / 0.0%	0.5570
2.0 (Clean Separation)	Oracle DML	0.0521	0.0421	0.0658	93.0% / 97.0%	0.4536
	Standard DML	8.4318	8.4508	8.4327	0.0% / 0.0%	4.9539
	DML + Z Controls (Ridge)	2.6307	2.6143	2.6345	0.0% / 0.0%	1.3505
	DML + Z Controls (GBM)	0.2066	0.1936	0.2426	72.0% / 72.0%	1.2144
	Regime FE DML (Hard)	0.0772	0.0634	0.1025	–	5.0238
	Spectral OR-DML (Ours)	0.0812	0.0720	0.1039	92.0% / 97.0%	0.4546
	Filtered OR-DML (Ours)	0.1015	0.0812	0.1306	92.0% / 94.0%	0.4556
4.0 (Isolated Regimes)	Oracle DML	0.0540	0.0477	0.0670	96.0% / 95.0%	0.4394
	Standard DML	8.4509	8.4611	8.4518	0.0% / 0.0%	4.9123
	DML + Z Controls (Ridge)	0.8479	0.8415	0.8534	0.0% / 0.0%	1.0945
	DML + Z Controls (GBM)	0.1006	0.0817	0.1248	89.0% / 89.0%	1.1889
	Regime FE DML (Hard)	0.0536	0.0461	0.0667	–	4.9960
	Spectral OR-DML (Ours)	0.0538	0.0462	0.0671	96.0% / 95.0%	0.4394
	Filtered OR-DML (Ours)	0.0538	0.0462	0.0671	96.0% / 95.0%	0.4394

Proof. Write the residualized treatment as $\tilde{T} = \Delta_T(S - \mathbb{E}[S | X, \hat{\gamma}]) + V$ and the residualized outcome as $\tilde{Y} = \theta\tilde{T} + \Delta_g(S - \mathbb{E}[S | X, \hat{\gamma}]) + U$. Taking the population residual regression coefficient gives $\theta + \text{Cov}(\tilde{T}, \Delta_g(S - \mathbb{E}[S | X, \hat{\gamma}])) / \text{Var}(\tilde{T})$, which yields the scalar identity; the vector form follows by the same covariance calculation. $\square$

Differentiating shows $|B|$ is non-monotone in $|\Delta_T|$ and is maximized at $|\Delta_T| = \sigma/\sqrt{v}$, with $\sup_{\Delta_T} |B| = |\Delta_g|\sqrt{v}/(2\sigma)$. This explains why empirical score conditioning alone cannot be treated as a monotone reliability certificate. The associated robustness value solves the identity for the smallest latent outcome shift capable of explaining the reported effect.

Audit qualifications. The posterior plug-in $\hat{v} = \mathbb{E}[\hat{\gamma}(1 - \hat{\gamma})]$ equals posterior uncertainty under correct calibration but need not equal realized residual-state error under misspecified emissions. In the committed misspecification experiment it is therefore treated as a lower-bound/optimistic diagnostic rather than an exact estimate. Fold-wise versus global HMM fitting is empirically nearly indistinguishable in the audited simulation, but this is evidence rather than a proof of independence. The isolated-switch approximation and learned-HMM rate are retained only with the assumptions and remainder qualifications documented in docs/V2_CHANGELOG.md; no stronger claim is imported into the main paper.